

Dynamic Response III — Time-Domain Specifications and the Effects of Zeros

FPE 8th ed., Sections 3.4 and 3.5

Audience: Fourth-year undergraduate mechanical engineering (ME 404)

Source: Franklin, Powell & Emami-Naeini, *Feedback Control of Dynamic Systems*, 8th ed., §§3.4–3.5. Worked examples and figure numbers are the book's. Additions of my own are marked **[beyond the book]**.

Prerequisites: L2 of this series (§3.3: pole locations, ζ , ω_n , σ , ω_d , the Fig. 3.18 geometry). The physical origin of zeros is in [From Physical Models to the Laplace Transform](#), cited as (0.0.8 §n).

Earlier-lecture shorthand: “0.0.8 §n” below refers to section n of the linked prerequisite, whose current filename is [modeling-and-dynamics_instructor.md](#).

Duration: 75 minutes.

Book figures: figure numbers refer to the source chapter at

references/documents/Franklin, Powell, Emami-Naeimi – 8th Ed./3 – Dynamic Response.pdf

Page cues below use the PDF viewer's 1-based page numbers (202 pages). Project the figures directly from this PDF; extracted image files are not supplied. **[beyond the book]** Classroom checks and teaching prompts are instructor additions. Demo paths are relative to [demos/](#); the demonstration index lists the scripts.

Notation:

Symbol	Meaning
t_r	rise time, from 10% to 90% of the final value
t_p	peak time
M_p	overshoot, as a fraction of the final value
t_s	settling time, to within 1% here (2% and 5% are also used)
$\sigma = \zeta\omega_n$	distance of the pole from the imaginary axis; positive for a stable pole
α	normalised zero (or extra-pole) location: the zero sits at $s = -\alpha\zeta\omega_n = -\alpha\sigma$

Part 0 — Planning

1. Teaching strategy

Last lecture ended with a picture and no numbers. Today supplies the numbers, and then immediately tells the truth about them.

The lecture has a clean two-act shape:

Act I (§3.4). Three formulas connect a step response to a pole location:

$$t_r \simeq \frac{1.8}{\omega_n}, \quad M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \quad t_s \simeq \frac{4.6}{\sigma}.$$

Inverted, they turn a specification into a *region* of the s -plane. That is the single most useful thing in the chapter for design.

Act II (§3.5). Every one of those formulas assumed two poles and no zeros. Real systems have zeros and extra poles. So: what do they change, by how much, and when can you ignore them?

The through-line to say out loud:

Act I gives you somewhere to put the poles. Act II tells you why the answer will not be quite what you asked for, and by how much.

Three moments to protect:

1. **§6** — Example 3.27, a specification drawn as a region. Everything in Chapter 5 lands in this picture.
2. **§9** — the derivative decomposition, $y = y_0 + \frac{1}{\alpha\zeta}\dot{y}_0$. It explains a LHP zero and a RHP zero with one equation.
3. **§11** — the Boeing 747 going down before it goes up. This is the moment nonminimum phase stops being vocabulary.

2. Learning objectives

By the end of this lecture students should be able to:

1. Define t_r , t_p , M_p and t_s off a step-response plot, and state the convention each definition depends on.
2. Derive $t_p = \pi/\omega_d$ and $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$ from the second-order step response.
3. Explain why $t_s \simeq 4.6/\sigma$ is an approximate decay-time rule, distinguish it from a conservative envelope bound, and explain why $t_r \simeq 1.8/\omega_n$ is a fit.
4. Convert a set of specifications into ω_n , ζ and σ bounds, and shade the allowable region of the s -plane (Example 3.27).
5. Give the first-order results: $M_p = 0$, $t_r = 2.2/\sigma$, $t_s = 4.6/\sigma$.
6. Explain how a zero near a pole reduces that mode's coefficient, using the cover-up formula (Eqs. 3.78–3.79).
7. Decompose the step response of a system with a zero into $y_0 + \frac{1}{\alpha\zeta}\dot{y}_0$, and use it to predict increased overshoot for a LHP zero and initial undershoot for a RHP zero.
8. State the factor-of-four rule for both zeros and extra poles, and the direction of each effect.
9. Recognise nonminimum-phase behaviour in a physical system and explain the mechanism (Example 3.30).
10. Judge when the second-order estimates may be used on a higher-order plant, and verify the judgement by simulation.

3. Suggested lecture flow (75 minutes)

Time	Topic	Section
0–4 min	Recap; today's question: how much overshoot, settled by when?	§4
4–10 min	The four definitions (Fig. 3.23)	§5.1
10–16 min	Rise time, and why 1.8	§5.2
16–26 min	Overshoot and peak time, derived	§5.3
26–32 min	Settling time and the envelope; the first-order case	§5.4–§5.5
32–42 min	Design synthesis; Example 3.27; the region	§6
42–52 min	Zeros: coefficient modification, the normalised family, Fig. 3.29	§7–§8
52–60 min	The derivative decomposition; RHP zeros	§9–§10
60–68 min	Example 3.28, Example 3.29; the Boeing 747	§10.2–§11
68–75 min	Extra poles; the four conclusions; closing	§12–§13

Prepare as slides: Figs. 3.23, 3.24, 3.25, 3.26 (definitions and regions) and Figs. 3.27–3.38 (the zero and extra-pole families). There are many figures in §3.5 and almost no algebra; resist deriving what a picture already settles. Reserve the board for the M_p derivation (§5.3), the Example 3.27 arithmetic (§6), and Eq. (3.81) (§9).

If you are short of time, drop Example 3.29 (complex zeros near lightly damped poles) and cover the extra-pole material from the summary box alone. Protect §6 and §11.

Runnable demonstrations

```
cd demos
uv run python ch3/l3_demo2_spec_regions.py --show
uv run python ch3/l3_demo4_extra_pole_aircraft.py --show
```

Run Demo 2 and Demo 4 live. Demo 2 shows a pole pair placed exactly on the specification boundary and then failing the specification, which is the honest version of §3.4. Demo 4 is the aeroplane. Demos 1 and 3 are prepared figures.

Teaching scope

Use the demos for the selected live examples; the accompanying checks also work on the board. For a 75-minute session, select the protected examples in §1; assign the remaining worked examples and optional computations as follow-up reading. Do not try to present every figure and derivation live.

Part I — The lecture

4. Opening

Instructor script

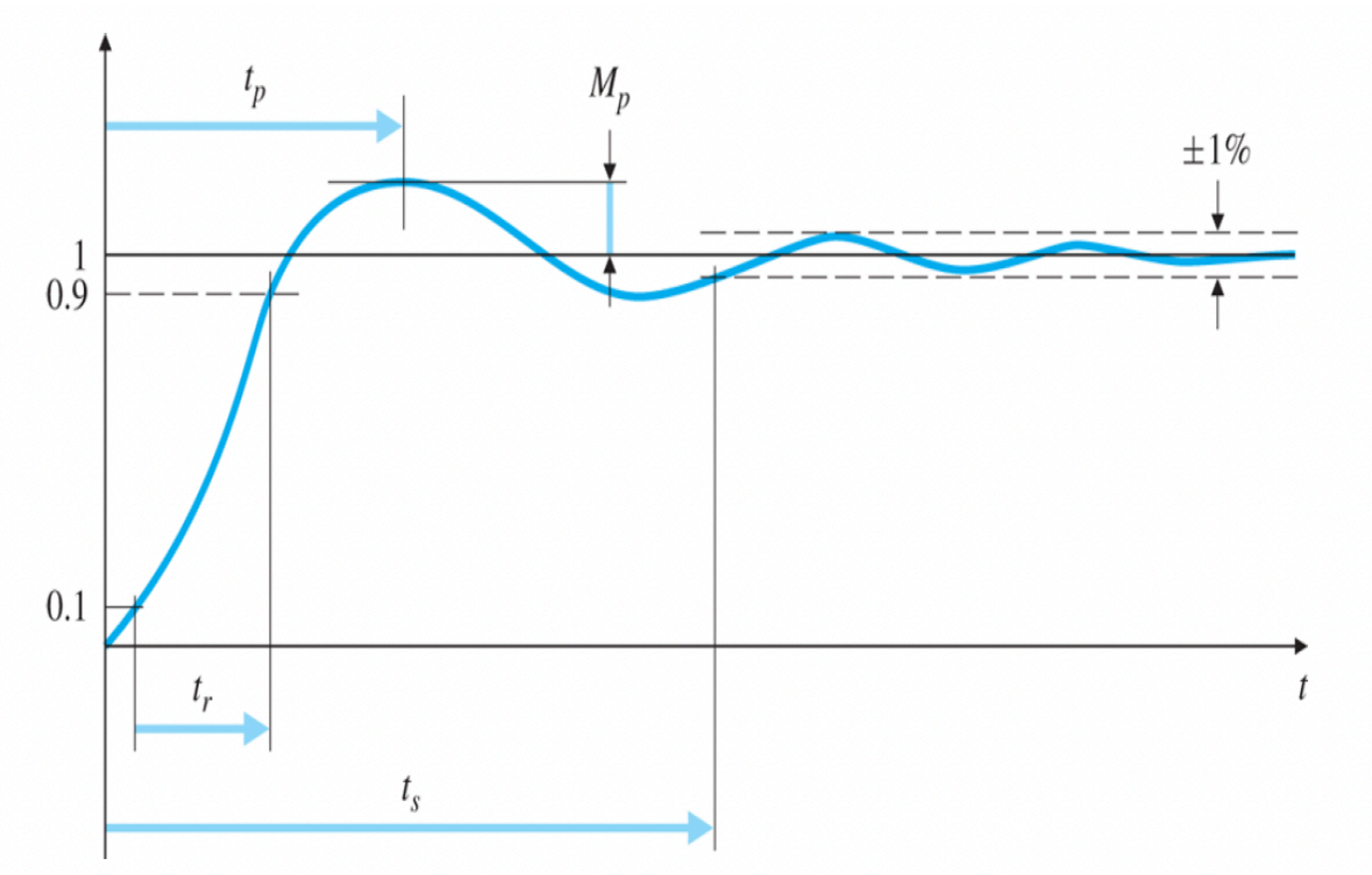
Last time we learned to look at a pole and say "fast", "slow", "rings", "unstable". That is a real skill, and it is not enough to design with.

A customer does not ask for a well-damped pair. They ask for: reaches the new setpoint in under half a second, overshoots by no more than ten percent, and is settled within three.

So today has two jobs. First, convert those three numbers into a place to put the poles. Second — and this is the part the book is unusually honest about — find out how much to trust the conversion.

5. The four specifications (§3.4)

5.1 Definitions



[FIG 3.23] — PDF p. 106 (slide; leave it up for the whole of §5)

Quantity	Definition
Rise time t_r	time to get to the vicinity of the new setpoint; the book uses 10% to 90%
Settling time t_s	time for the transient to decay and stay inside a band; 1% here
Overshoot M_p	for the positive unit-step responses here, the maximum excess over the final value, divided by the final value
Peak time t_p	time at which that maximum occurs

Say out loud

Two of these depend on a convention someone chose. Rise time could be 0% to 100%, or 5% to 95%. Settling could be to 1%, 2% or 5%. When you read a specification from a customer or a datasheet, **find out which convention it uses** before you compare it with anything.

5.2 Rise time

The book's argument is a fit, not a derivation: look at the family of step responses in Fig. 3.19(b), take $\zeta = 0.5$ as an average, measure, and get $\omega_n t_r \approx 1.8$. Hence

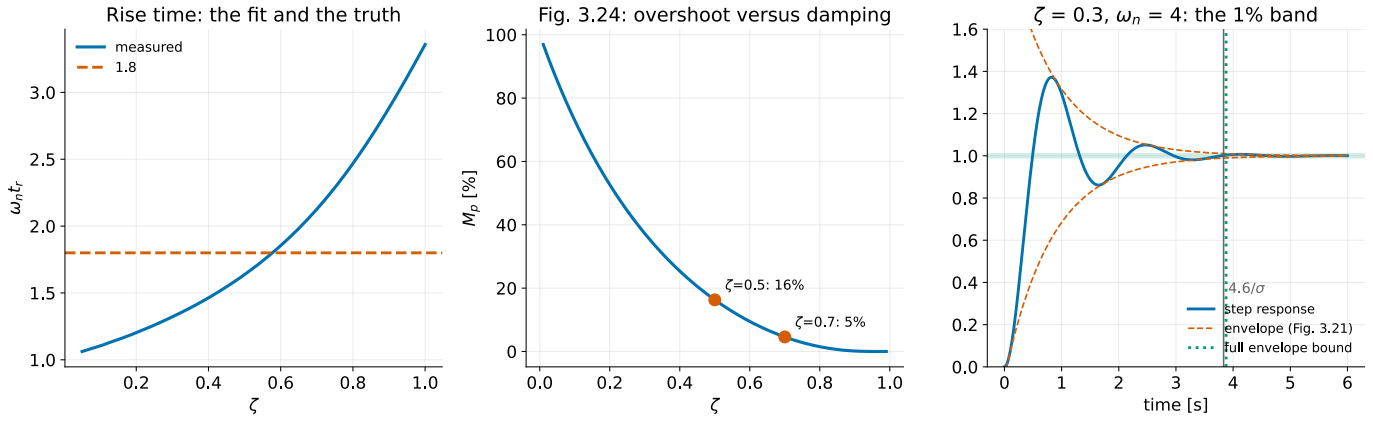
$$t_r \simeq \frac{1.8}{\omega_n}$$

(3.68)

[DEMO 1] — `ch3/l3_demo1_step_specs.py` (slide)

Teaching check [beyond the book]: Compute the 10% and 90% crossings of the standard step response. The products $\omega_n t_r$ are approximately 1.10, 1.64, 2.13, and 3.36 at $\zeta = 0.1, 0.5, 0.7, 1$, respectively. The value 1.8 is attained near $\zeta = 0.58$; it is a rough fit, not a bound.

Time-domain specifications: exact peaks, approximate rise and settling



[beyond the book] That last sentence has a consequence students should hear now rather than discover in a lab: a pole pair placed exactly on the ω_n boundary can *fail* the rise-time requirement if its damping is high. Demo 2 shows exactly that case.

5.3 Overshoot and peak time — derive this one

For zero initial state, unit DC gain, $\omega_n > 0$, and $0 < \zeta < 1$, the step response of the standard second-order system is

$$y(t) = 1 - e^{-\sigma t} \left(\cos \omega_d t + \frac{\sigma}{\omega_d} \sin \omega_d t \right) \quad (3.69)$$

(derived by partial fractions in L2 §10.2). The identity $A \sin \alpha + B \cos \alpha = C \cos(\alpha - \beta)$ compresses it. Here the bracket has $B = 1$ and $A = \sigma/\omega_d = \zeta/\sqrt{1-\zeta^2}$, so

$$C = \sqrt{1 + \frac{\zeta^2}{1-\zeta^2}} = \frac{1}{\sqrt{1-\zeta^2}}, \quad \cos \beta = \frac{B}{C} = \sqrt{1-\zeta^2}, \quad \sin \beta = \frac{A}{C} = \zeta.$$

Hence $\beta = \sin^{-1} \zeta$, the same angle θ as in the Fig. 3.18 pole geometry, and

$$y(t) = 1 - \frac{e^{-\sigma t}}{\sqrt{1-\zeta^2}} \cos(\omega_d t - \beta), \quad \beta = \sin^{-1} \zeta \quad (3.70)$$

Source correction [beyond the book]: Eq. (3.70) on PDF p. 108 prints $\sqrt{1-\zeta}$ in the denominator; it should be $\sqrt{1-\zeta^2}$, as used here and in the envelope bound in §5.4.

Differentiate Eq. (3.69) with the product rule. The derivative of $-e^{-\sigma t}$ is $+\sigma e^{-\sigma t}$, and the bracket differentiates to $-\omega_d \sin \omega_d t + \sigma \cos \omega_d t$:

$$\dot{y}(t) = \sigma e^{-\sigma t} \left(\cos \omega_d t + \frac{\sigma}{\omega_d} \sin \omega_d t \right) - e^{-\sigma t} (-\omega_d \sin \omega_d t + \sigma \cos \omega_d t).$$

The two cosine terms, $+\sigma \cos$ and $-\sigma \cos$, cancel. That is the collapse. What remains is

$$\dot{y}(t) = e^{-\sigma t} \left(\frac{\sigma^2}{\omega_d} + \omega_d \right) \sin \omega_d t = \frac{\omega_n^2}{\omega_d} e^{-\sigma t} \sin \omega_d t,$$

using $\sigma^2 + \omega_d^2 = \omega_n^2$. This is the impulse response, Eq. (3.66), as it must be.

Everything except $\sin \omega_d t$ is strictly positive, so nonzero-time extrema occur at $\omega_d t = k\pi$. Odd k give maxima and even k minima. The first peak is at $\omega_d t_p = \pi$:

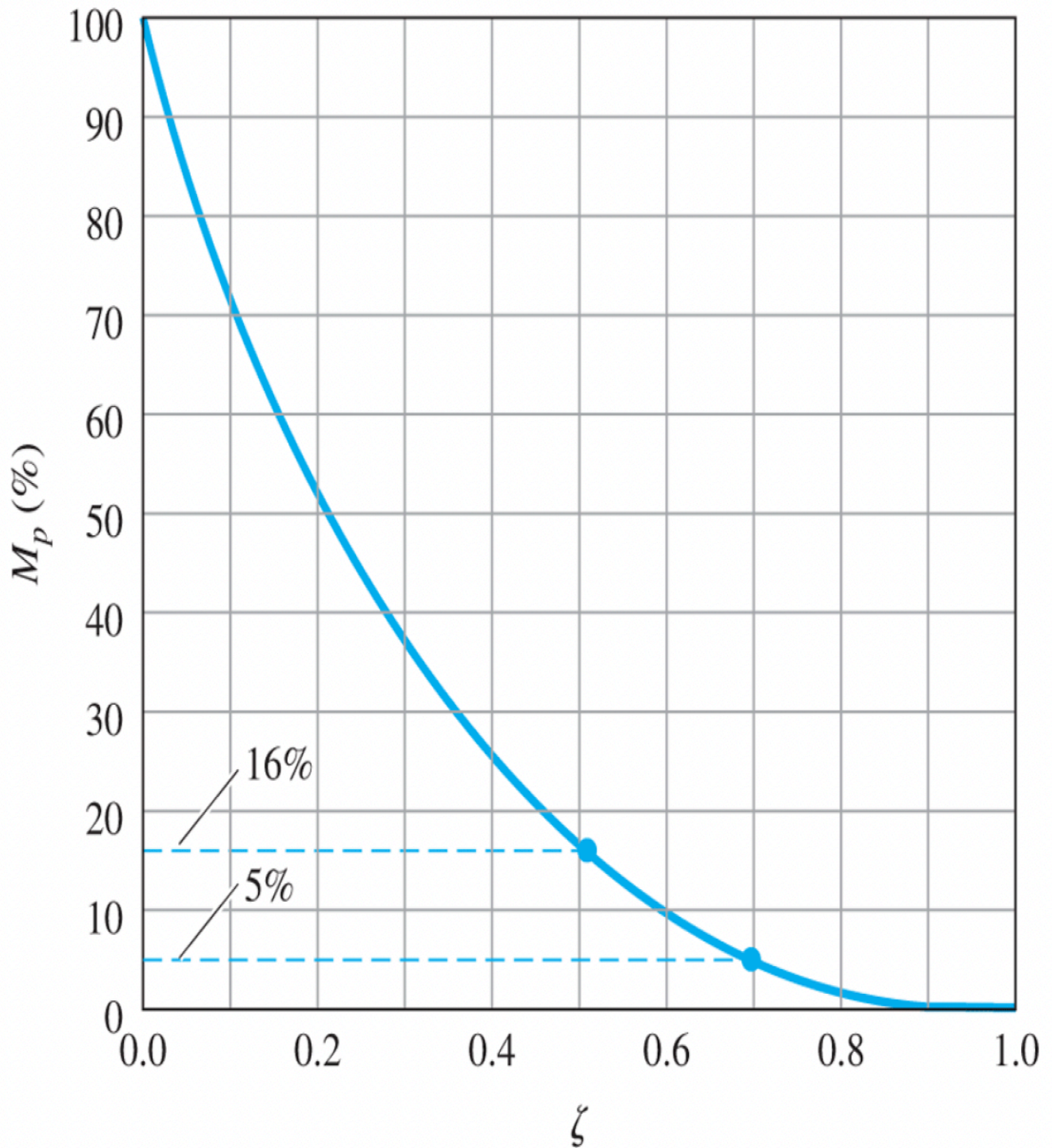
$$t_p = \frac{\pi}{\omega_d} \quad (3.71)$$

Substituting $\omega_d t_p = \pi$ into Eq. (3.69), $\cos \pi = -1$ and $\sin \pi = 0$:

$$y(t_p) = 1 - e^{-\sigma\pi/\omega_d} (-1 + 0) = 1 + e^{-\sigma\pi/\omega_d}.$$

The final value is 1, so the overshoot is $M_p = e^{-\sigma\pi/\omega_d}$. Finally, $\sigma/\omega_d = \zeta\omega_n/(\omega_n\sqrt{1-\zeta^2}) = \zeta/\sqrt{1-\zeta^2}$:

$$M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}, \quad 0 < \zeta < 1 \quad (3.72)$$



[FIG 3.24] — PDF p. 110 (slide)

At $\zeta = 0$, the same peak formula gives a 100% excursion above the equilibrium value, but the response never settles and has no final-value limit. For $\zeta \geq 1$, the zero-free standard step response is monotone: $M_p = 0$ and there is no finite overshoot peak time.

Two values to memorise, and they should be memorised. At $\zeta = 0.5$ the exponent is $\pi(0.5)/0.866 = 1.814$, and $e^{-1.814} = 0.163$. At $\zeta = 0.7$ it is $\pi(0.7)/0.714 = 3.079$, and $e^{-3.079} = 0.046$:

$$\zeta = 0.5 \Rightarrow M_p = 16\% \quad \zeta = 0.7 \Rightarrow M_p = 5\%$$

Ask the class

M_p depends on ζ alone. Not on ω_n . Why should that be?

Because ω_n only scales time. Two systems with the same ζ have the same step response shape, one played faster than the other. Overshoot is a property of the shape.

Algebraically, $\sigma t = \zeta(\omega_n t)$ and $\omega_d t = \sqrt{1 - \zeta^2}(\omega_n t)$. Eq. (3.69) is therefore a function of ζ and the normalised time $\omega_n t$ only. The peak *height* depends on ζ alone, and the peak *time* scales as $1/\omega_n$.

5.4 Settling time

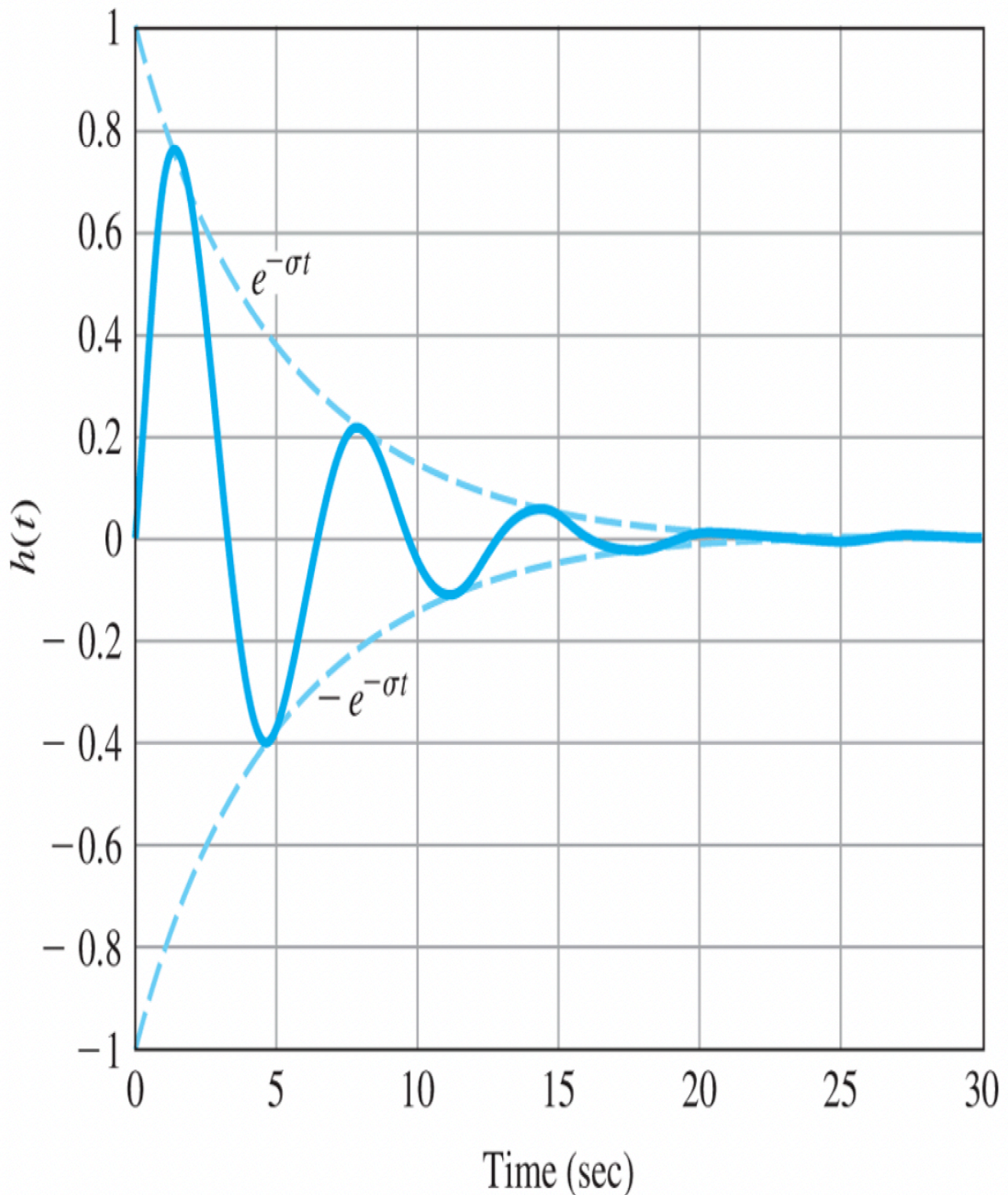
The textbook uses the exponential decay rate to estimate the 1% settling time. Dropping the oscillatory phase and the envelope prefactor gives

$$e^{-\sigma t_s} = 0.01 \implies \sigma t_s = -\ln 0.01 = \ln 100 = 4.605 \implies \boxed{t_s \simeq \frac{4.6}{\zeta \omega_n} = \frac{4.6}{\sigma}} \quad (3.73)$$

Qualification [beyond the book]: Eq. (3.70) gives the full bound

$$|y(t) - 1| \leq \frac{e^{-\sigma t}}{\sqrt{1 - \zeta^2}}, \quad t_{s,\text{env}} = \frac{-\ln(\epsilon \sqrt{1 - \zeta^2})}{\sigma}, \quad 0 < \zeta < 1.$$

The bound follows from Eq. (3.70) because $|\cos(\cdot)| \leq 1$. To get $t_{s,\text{env}}$, set the bound equal to ϵ and take logs: $e^{-\sigma t} = \epsilon \sqrt{1 - \zeta^2}$, so $\sigma t = -\ln(\epsilon \sqrt{1 - \zeta^2})$. For a tolerance $\epsilon = 0.01$, the response stays in the band once $t \geq t_{s,\text{env}}$. The actual settling time is the last band crossing and can be earlier. The shorter rule $4.6/\sigma$ is **not a guaranteed upper bound**. Near critical damping the sinusoidal envelope bound is very loose; use the actual response. At $\zeta = 1$, the step response is $1 - (1 + \omega_n t)e^{-\omega_n t}$ (L2 §10.2), which approaches 1 monotonically from below. Setting the error $(1 + x)e^{-x} = 0.01$ with $x = \omega_n t$ gives a transcendental equation. Its root is $x \approx 6.64$ (Newton's method from $x = 6$ gives 6.49, 6.63, 6.64; or use `fsolve`), so $t_s \approx 6.64/\omega_n$, not $4.6/\omega_n$. The factor $(1 + x)$ is what the pure-exponential rule misses.



[FIG 3.21] — PDF p. 99 (recall from L2: the envelope)

[DEMO 1, continued] — right-hand panel

Teaching check [beyond the book]: At $\zeta = 0.7$, $\omega_n = 4$ rad/s, the actual 1% settling time is approximately 1.644 s, slightly later than $4.6/\sigma = 1.643$ s. The corrected envelope bound above is approximately 1.765 s. Arithmetic: $\sigma = 0.7 \times 4 = 2.8$ s⁻¹, $4.6/2.8 = 1.643$ s, $\sqrt{1 - 0.49} = 0.714$, and $-\ln(0.00714)/2.8 = 4.942/2.8 = 1.765$ s. Thus the textbook estimate is not guaranteed conservative.

5.5 The first-order case

For $H(s) = \frac{\sigma}{s + \sigma}$ the step response is

$$y(t) = (1 - e^{-\sigma t}) 1(t) \quad (3.77)$$

(the step response of L2 §8.2, scaled to unit DC gain). Each specification follows by solving $y(t) = \text{level}$:

- **Overshoot.** $\dot{y} = \sigma e^{-\sigma t} > 0$, so y rises monotonically to 1 and never exceeds it: $M_p = 0$.
- **Rise time.** $1 - e^{-\sigma t} = p$ gives $t = -\ln(1 - p)/\sigma$. So $t_{10} = -\ln 0.9/\sigma = 0.105/\sigma$ and $t_{90} = -\ln 0.1/\sigma = 2.303/\sigma$. The difference is $t_r = (\ln 10 - \ln \frac{10}{9})/\sigma = \ln 9/\sigma = 2.197/\sigma$.
- **Settling time.** The error is $e^{-\sigma t}$ exactly, with no oscillation and no prefactor. It reaches 0.01 at $t_s = \ln 100/\sigma = 4.605/\sigma$.

$$M_p = 0, \quad t_r = \frac{\ln 9}{\sigma} \approx \frac{2.2}{\sigma}, \quad t_s = \frac{\ln 100}{\sigma} \approx \frac{4.6}{\sigma}, \quad \tau = \frac{1}{\sigma}$$

Note that t_r here is *derived exactly*, unlike the second-order 1.8. Point that out; it explains why the second-order constant is different and approximate.

6. Design synthesis: specifications become a region

Invert the three formulas to obtain an **approximate design region**. Here t_r, t_s, M_p denote the specified upper limits; only the overshoot constraint is exact for the standard underdamped pair:

$$\omega_n \geq \frac{1.8}{t_r} \quad \zeta \geq \zeta(M_p) \quad \sigma \geq \frac{4.6}{t_s} \quad (3.74\text{--}3.76)$$

Each is a geometric constraint:

For $0 < M_p < 1$, the exact damping bound is

$$\zeta \geq \frac{-\ln M_p}{\sqrt{\pi^2 + (\ln M_p)^2}}.$$

Derivation. Take logs of Eq. (3.72) and write $L = -\ln M_p > 0$:

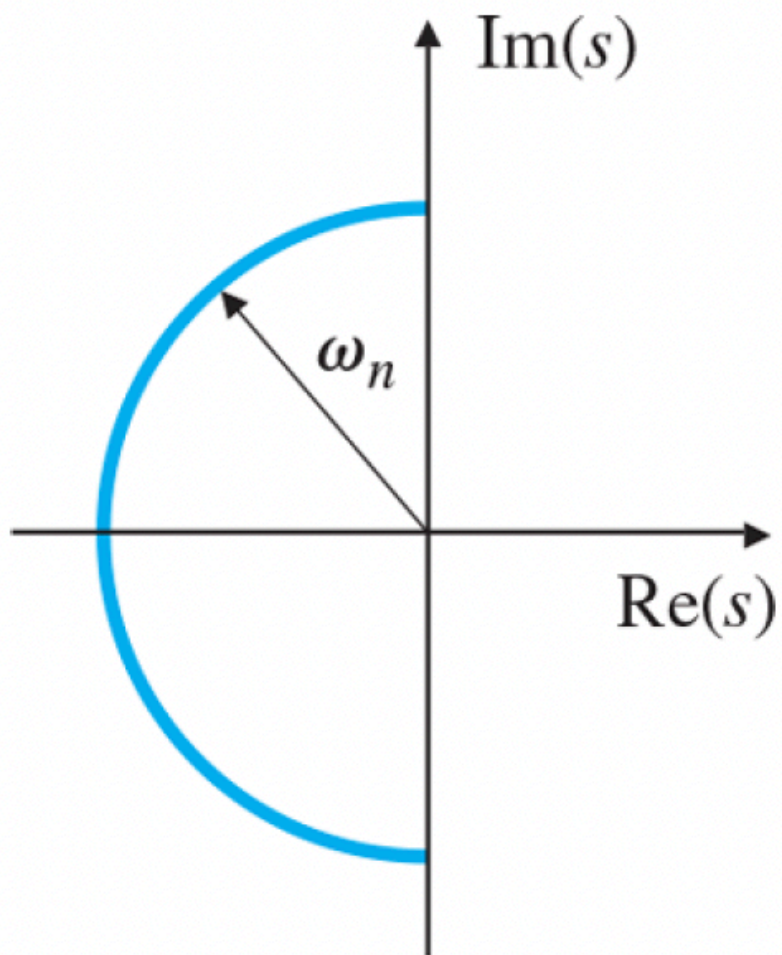
$$\frac{\pi\zeta}{\sqrt{1-\zeta^2}} = L \implies \pi^2\zeta^2 = L^2(1-\zeta^2) \implies \zeta^2(\pi^2 + L^2) = L^2 \implies \zeta = \frac{L}{\sqrt{\pi^2 + L^2}}.$$

Squaring is safe because both sides are positive for $0 < \zeta < 1$. M_p decreases as ζ increases, so " $M_p \leq \text{spec}$ " becomes " $\zeta \geq \text{this value}$ ".

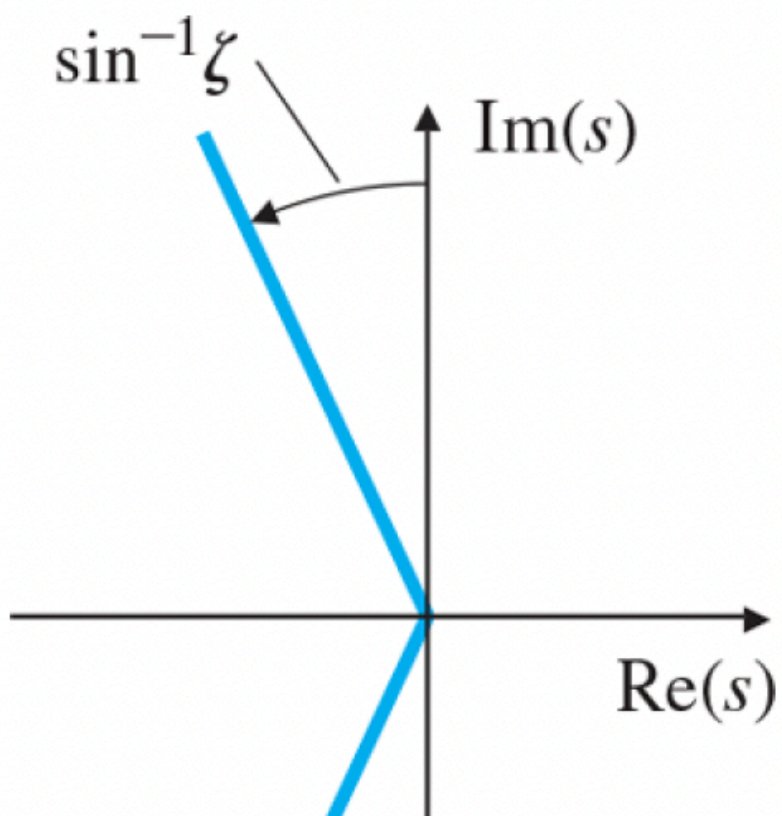
The other two bounds follow directly from Eqs. (3.68) and (3.73). $t_r \simeq 1.8/\omega_n \leq t_r^{\text{spec}}$ gives $\omega_n \geq 1.8/t_r^{\text{spec}}$, and $t_s \simeq 4.6/\sigma \leq t_s^{\text{spec}}$ gives $\sigma \geq 4.6/t_s^{\text{spec}}$.

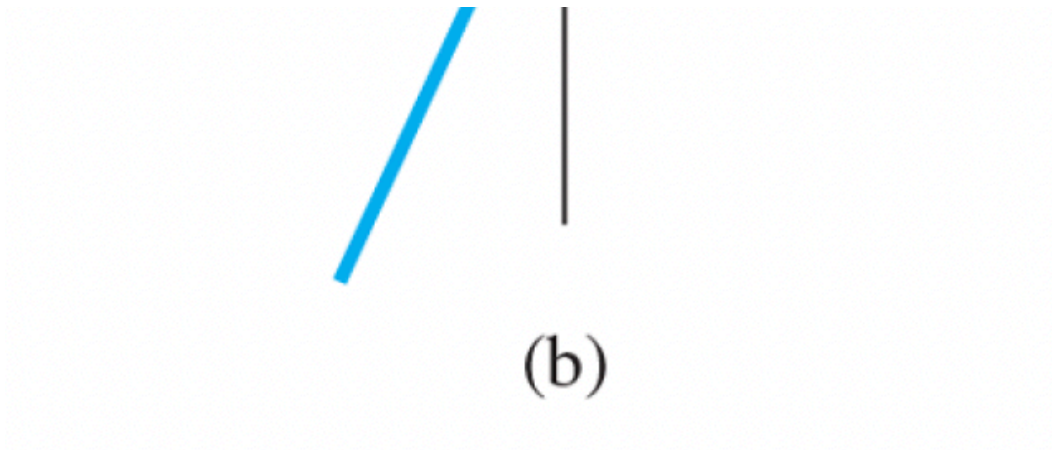
Why each is the stated shape (the L2 §10.1 geometry). ω_n is the pole's distance from the origin, so a lower bound on it excludes a disc. $\zeta = \sin \theta$ with θ measured from the imaginary axis, so a lower bound on ζ is a lower bound on θ : a wedge about the negative real axis. σ is the distance from the imaginary axis, so a lower bound on it is a half-plane to the left of a vertical line.

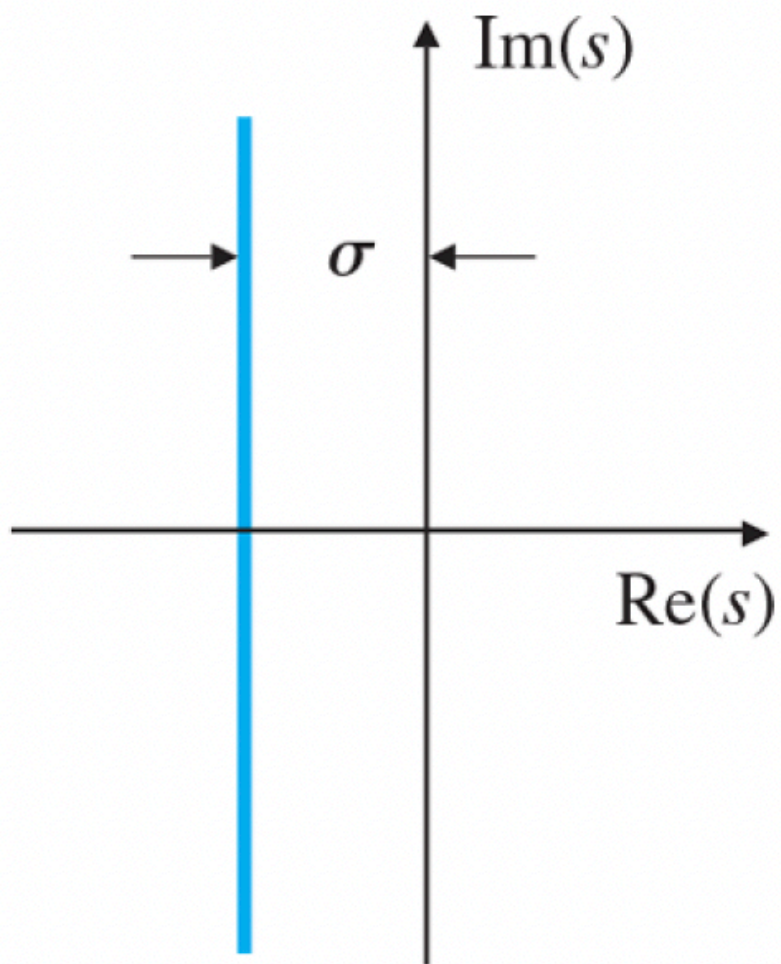
Bound	Region in the s -plane
$\omega_n \geq$	outside a circle of that radius
$\zeta \geq$	inside a wedge, measured by the angle $\theta = \sin^{-1} \zeta$ from the imaginary axis
$\sigma \geq$	to the left of a vertical line



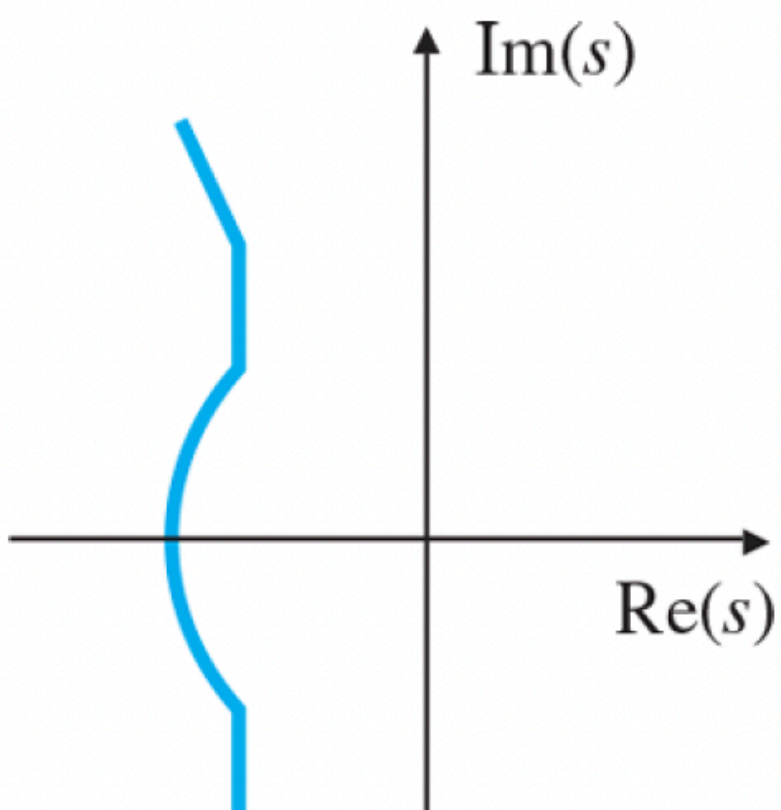
(a)







(c)





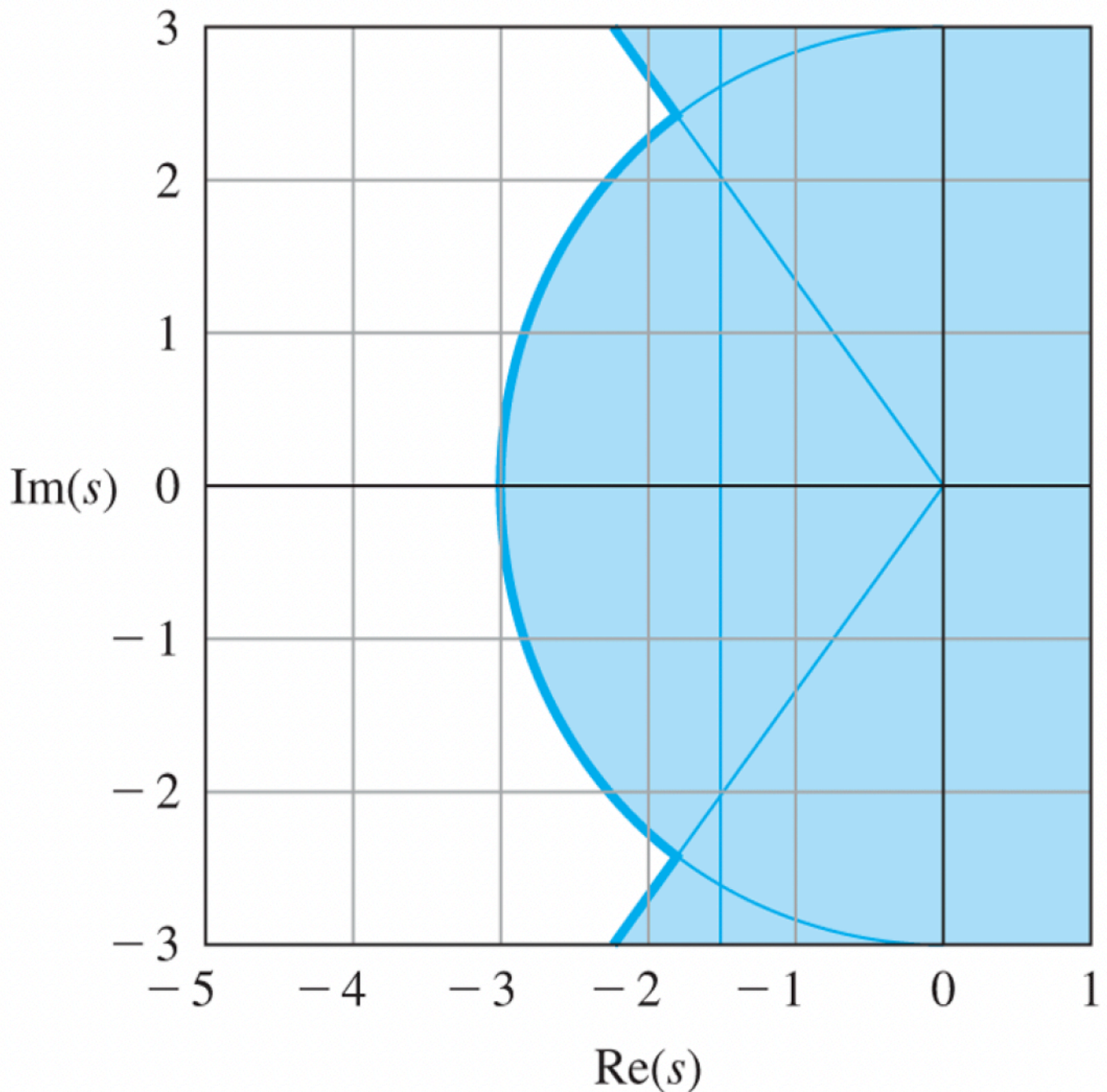
(d)

[FIG 3.25] — PDF p. 113, PDF p. 113, PDF p. 114, PDF p. 114 (*slide: the three regions and their intersection*) (a) rise-time, (b) overshoot, (c) settling time, (d) composite

6.1 Example 3.27

Requirements: $t_r \leq 0.6$ s, $M_p \leq 10\%$, $t_s \leq 3$ s. On the board:

$$\omega_n \geq \frac{1.8}{0.6} = 3 \text{ rad/s}, \quad \zeta \geq 0.6, \quad \sigma \geq \frac{4.6}{3} = 1.53 \text{ s}^{-1}.$$



[FIG 3.26] — PDF p. 117 (slide)

Careful: the shading in the book's Fig. 3.26 covers the region that is **excluded**; the allowable set is to the left of the solid boundary, as the text says. Say which is which before a student copies the picture into an exam answer.

The book's closing observation is worth the extra thirty seconds: any pole meeting the ω_n and ζ bounds automatically satisfies the σ bound, since $\sigma = \zeta\omega_n \geq 0.6 \times 3 = 1.8 > 1.53$. **The settling requirement is not binding here.** Recognising which specification is actually driving the design is a skill worth naming.

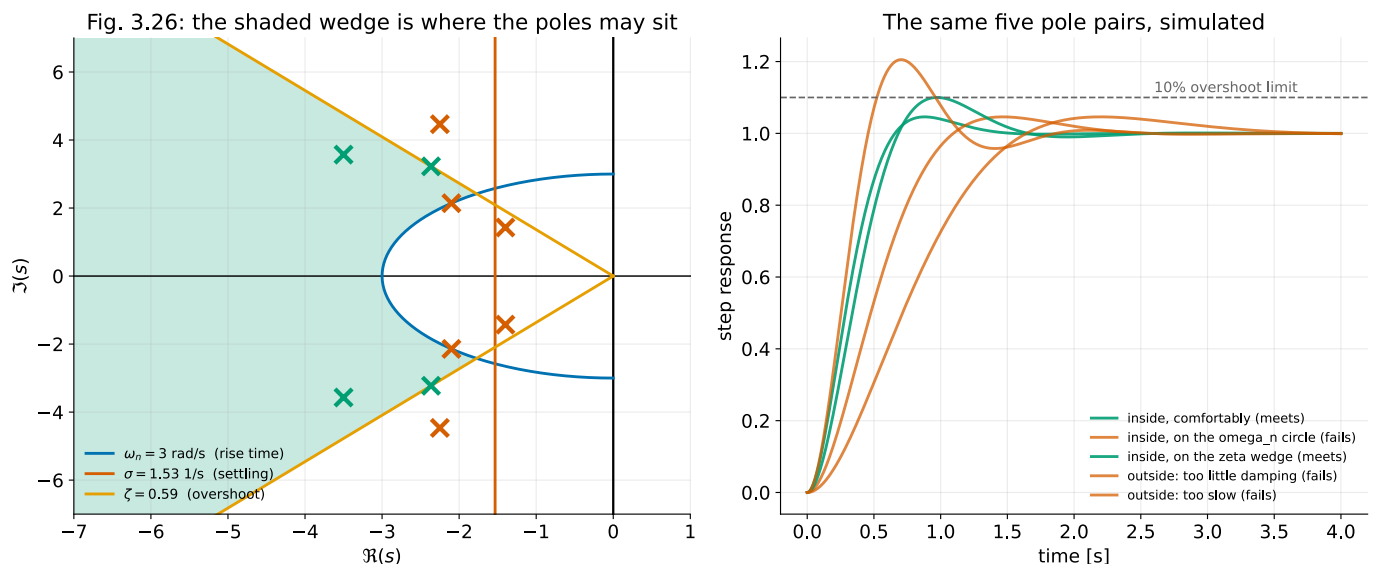
Two small corrections to make in passing [beyond the book]:

- Inverting $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$ at 10% gives $\zeta = 0.591$: $L = -\ln 0.1 = 2.303$, so $\zeta = 2.303/\sqrt{9.870 + 5.302} = 2.303/3.895 = 0.591$. The corresponding wedge half-angle is $\theta = \sin^{-1} 0.591 = 36.2^\circ$. The book reads 0.6 off Fig. 3.24, which is the same number rounded, but students should know the bound is exact and computable.
- The book prints " $\sigma \geq 4.6/3 = 1.5$ sec". The value is right and the unit is not: σ is a rate, in inverse seconds.

[DEMO 2] — [ch3/l3_demo2_spec_regions.py](#) (live, ~10 s — the most important demo of this lecture)

Teaching check [beyond the book]: Place the poles at $\zeta = 0.7$, $\omega_n = 3$ rad/s. They satisfy the approximate design region: $\omega_n = 3 \geq 3$, $\zeta = 0.7 \geq 0.591$, and $\sigma = 2.1 \geq 1.53$. Yet the actual 10–90% rise time is about 0.709 s, which is $\omega_n t_r \approx 2.13$ from §5.2 divided by $\omega_n = 3$, exceeding 0.6 s. The overshoot constraint is exact for the standard pair; the rise-time circle and settling-time line require verification.

Example 3.27: specifications drawn in the s-plane, then verified



7. Zeros, first pass: they reweight the modes (§3.5)

Instructor script

Everything so far assumed two poles and no zeros. Now let us add a zero and see what breaks.

Start with the cover-up formula from Lecture 1. The coefficient of the mode at p_1 is the whole transfer function, with the $(s - p_1)$ factor removed, evaluated at $s = p_1$.

So: if there is a zero *near* p_1 , that evaluation is near zero, and the coefficient is small. A zero near a pole switches that mode off.

The book's pair of examples, normalised to the same DC gain:

$$H_1(s) = \frac{2}{(s+1)(s+2)} = \frac{2}{s+1} - \frac{2}{s+2} \quad (3.78)$$

$$H_2(s) = \frac{2(s+1.1)}{1.1(s+1)(s+2)} = \frac{2/11}{s+1} + \frac{18/11}{s+2} \quad (3.79)$$

The residues, by cover-up. Both transfer functions have $H(0) = 1$: $2/(1 \cdot 2) = 1$ and $2(1.1)/(1.1 \cdot 1 \cdot 2) = 1$. The 1.1 in the denominator of H_2 is there to make that true.

$$H_1 : \quad \frac{2}{s+2} \Big|_{s=-1} = \frac{2}{1} = 2, \quad \frac{2}{s+1} \Big|_{s=-2} = \frac{2}{-1} = -2.$$

$$H_2 : \quad \frac{2(s+1.1)}{1.1(s+2)} \Big|_{s=-1} = \frac{2(0.1)}{1.1(1)} = \frac{0.2}{1.1} = \frac{2}{11}, \quad \frac{2(s+1.1)}{1.1(s+1)} \Big|_{s=-2} = \frac{2(-0.9)}{1.1(-1)} = \frac{1.8}{1.1} = \frac{18}{11}.$$

The factor $(s + 1.1)$ evaluated at the pole $s = -1$ is 0.1. That small number is the whole effect.

the coefficient of e^{-t} fell from 2 to $2/11 \approx 0.18$

because the zero at -1.1 nearly cancels the pole at -1 . Put the zero exactly at -1 and the term vanishes entirely.

Ask the class

If a zero can switch a mode off, is a pole-zero cancellation a good way to get rid of a mode you dislike?

Hold the answer. It is Example 3.29 in twenty minutes, and Lecture 4 in a week. The short version: it removes the mode from *this* transfer function only.

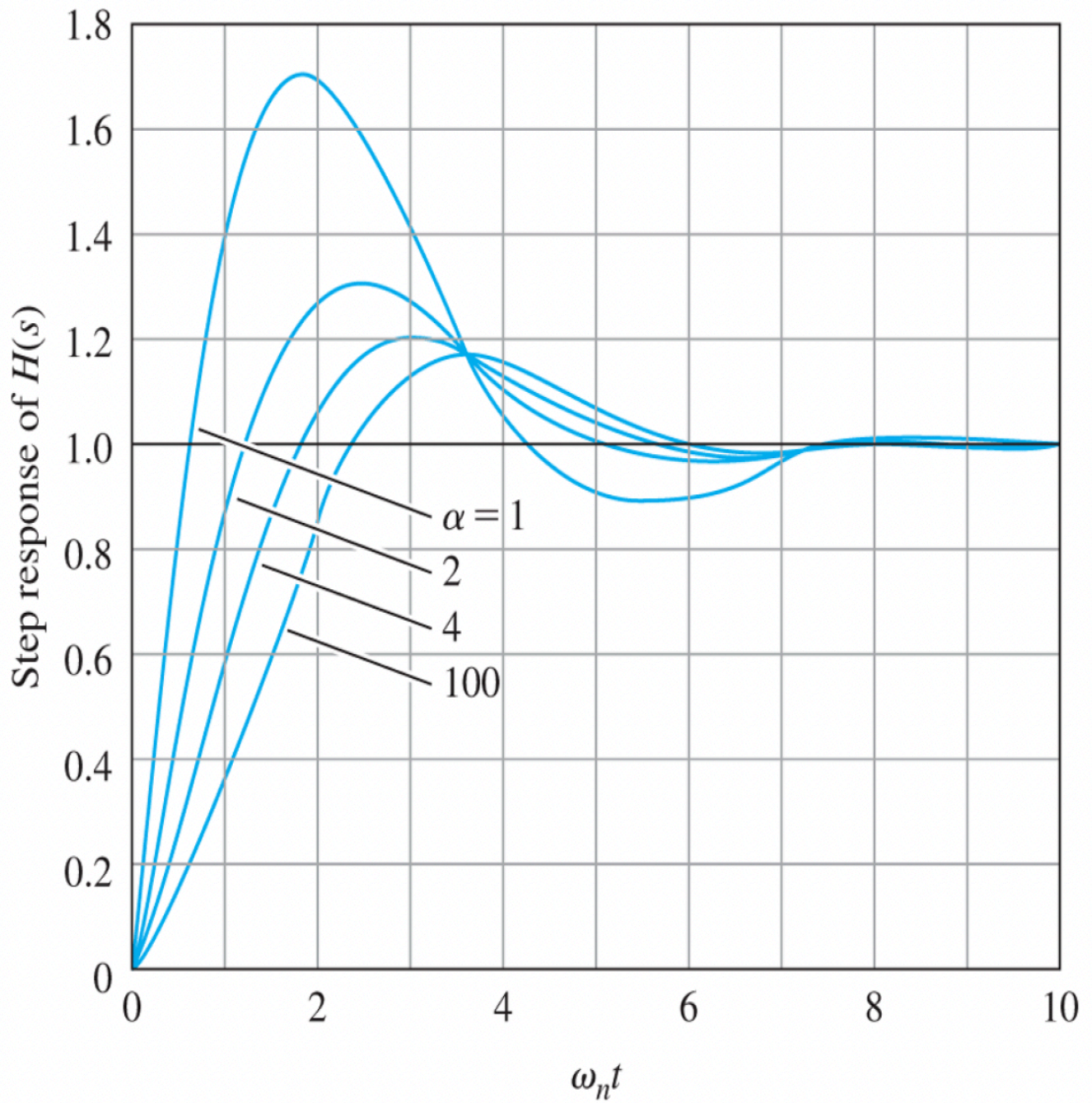
8. The normalised family

To study zeros systematically, put the zero at $s = -\alpha\zeta\omega_n = -\alpha\sigma$ and normalise:

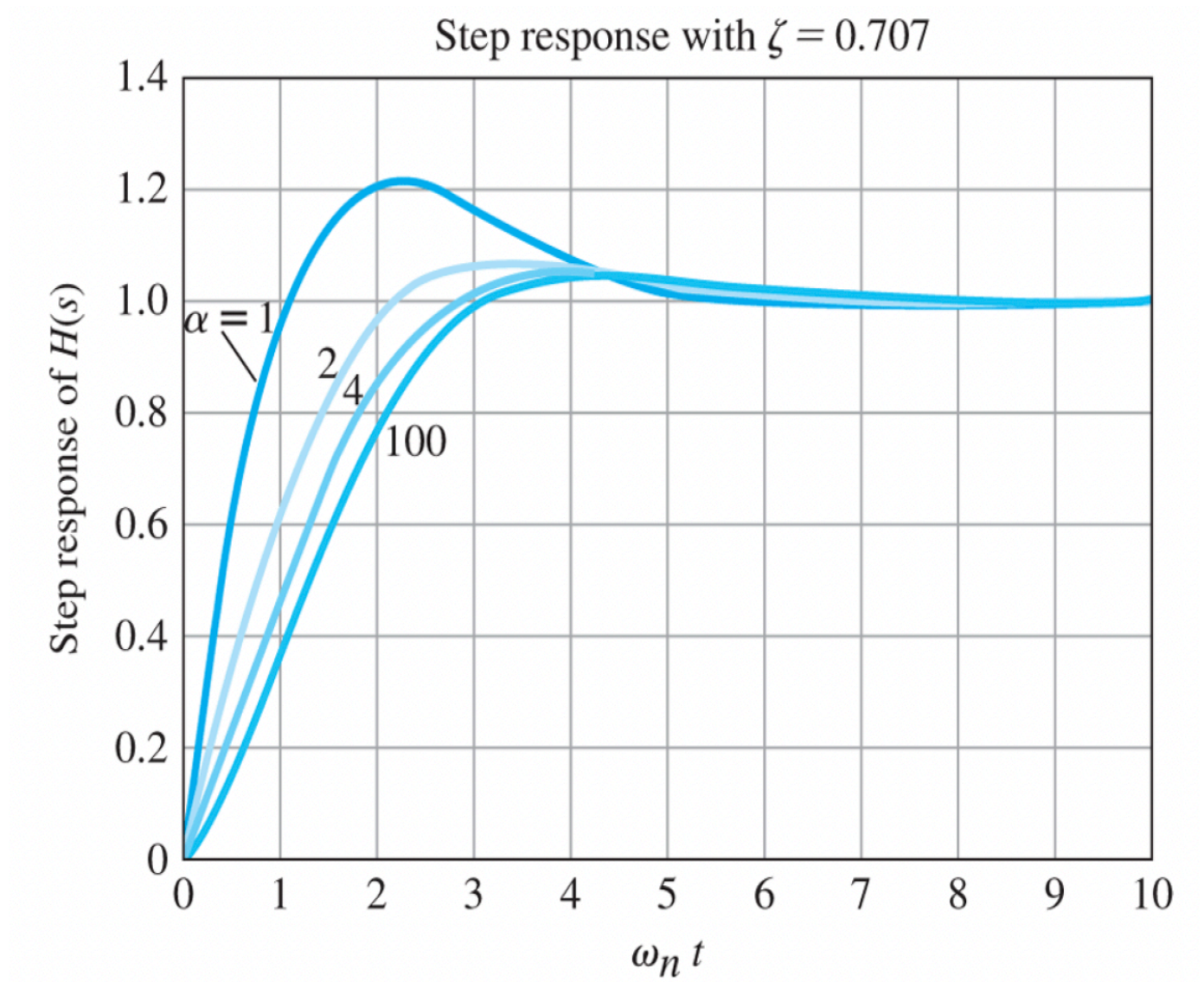
$$H(s) = \frac{(s/\alpha\zeta\omega_n) + 1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

(3.80)

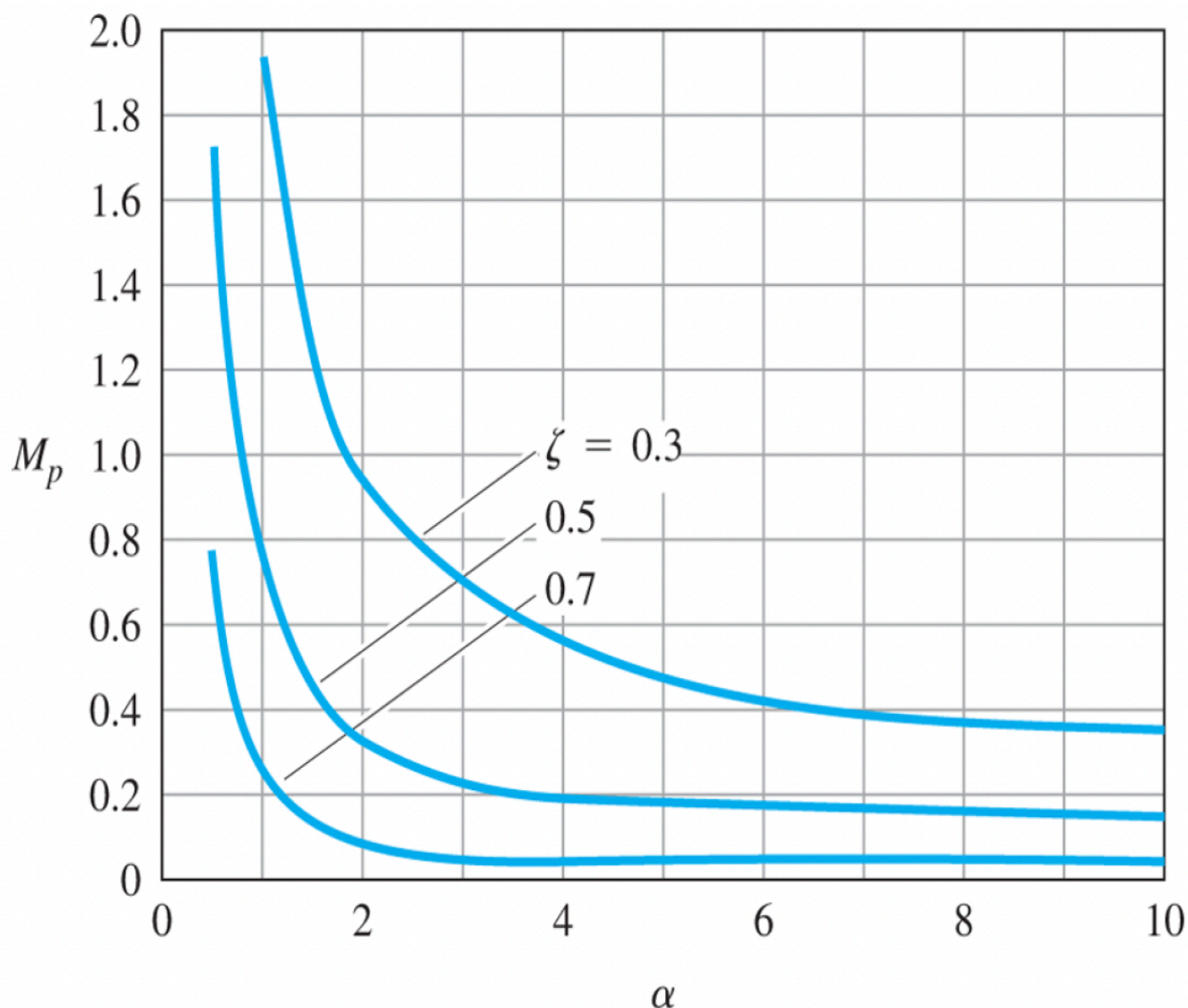
For $\alpha > 0$, read α as the ratio of the zero's distance from the imaginary axis to the pair's distance σ . Large α means a relatively distant LHP zero. $\alpha = 1$ means equal real parts, not coincidence with either complex pole. Negative α puts the zero in the RHP. Assume $0 < \zeta < 1$ in this family.



[FIG 3.27] — PDF p. 120 ($\zeta = 0.5$)



[FIG 3.28] — PDF p. 121 ($\zeta = 0.707$)



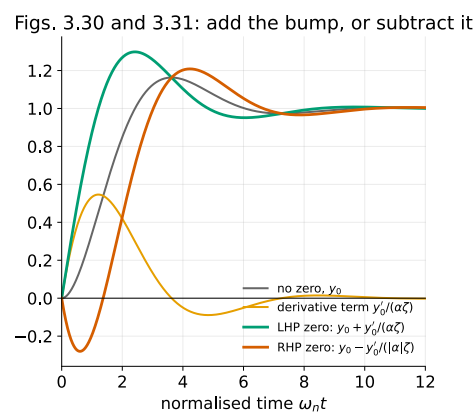
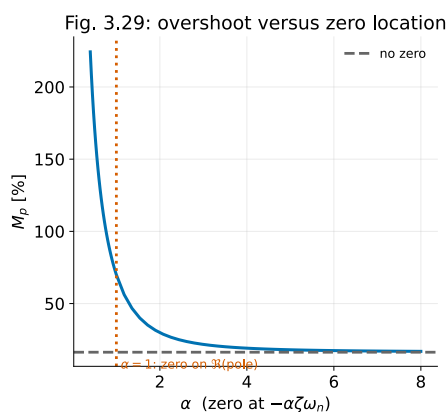
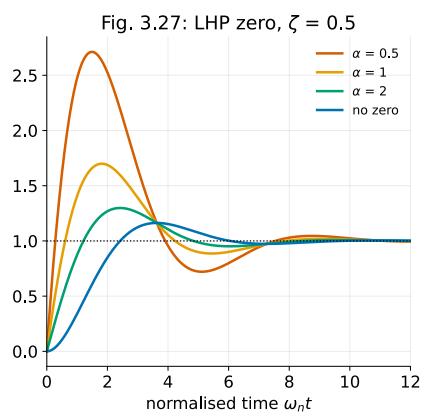
[FIG 3.29] — PDF p. 122 (overshoot versus α)

Show: Fig. 3.27, then Fig. 3.29 immediately after. **Say:** "In this family the LHP zero raises overshoot and quickens the rise. It leaves the pole decay rates unchanged, but changing modal amplitudes can still change settling time. Its effect diminishes with distance; decide whether it is negligible using the actual specification."

[DEMO 3] — [ch3/l3_demo3_zeros.py](#) (slide)

Teaching check [beyond the book]: For $\zeta = 0.5$, compare the zero-free overshoot (16.3%) with $\alpha = 4, 2, 1, 0.5$: approximately 19.1%, 29.8%, 69.9%, and 171%. A nearby zero can change the response substantially even though the poles are fixed.

Zeros do not move the poles; they reweight what the poles produce



The factor-of-four rule is the book's heuristic for when a real LHP zero is likely to matter. It is not an error bound: at $\alpha = 4$, overshoot still rises from 16.3% to 19.1% in this example.

9. Why: the derivative decomposition

This is the explanation that makes both the LHP and the RHP case obvious, so give it the board.

Set $\omega_n = 1$. Eq. (3.80) becomes

$$H(s) = \frac{\frac{s}{\alpha\zeta} + 1}{s^2 + 2\zeta s + 1}.$$

The numerator is a sum of two terms, so split the fraction into two terms over the same denominator:

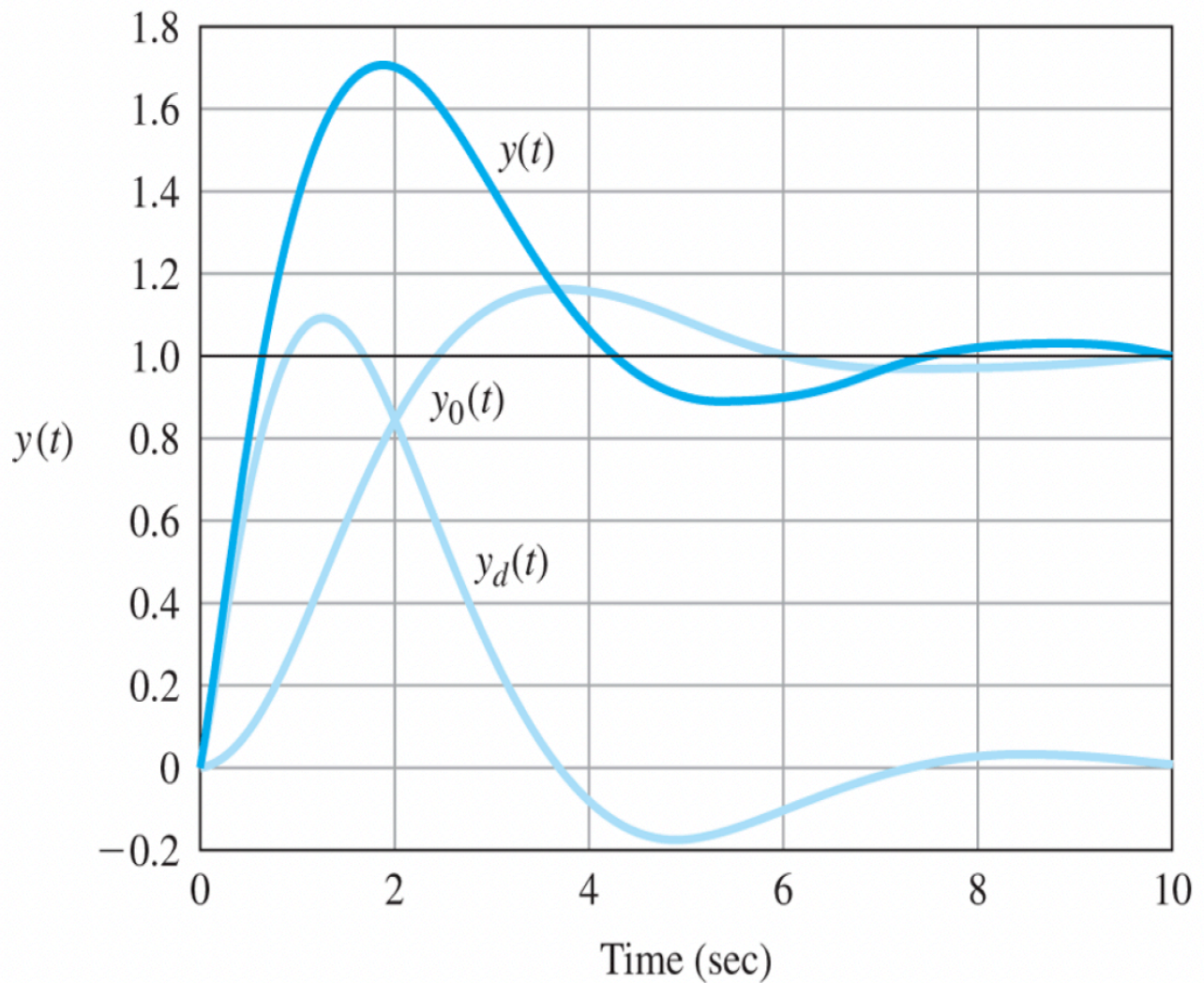
$$H(s) = \underbrace{\frac{1}{s^2 + 2\zeta s + 1}}_{H_0(s)} + \frac{1}{\alpha\zeta} \underbrace{\frac{s}{s^2 + 2\zeta s + 1}}_{sH_0(s)} \quad (3.81)$$

The second term is a constant times s times the first. With a step input, $Y = H/s$, so $Y = Y_0 + \frac{1}{\alpha\zeta} sY_0$ with $Y_0 = H_0/s$. By property 5 of L1 §9, $\mathcal{L}\{\dot{y}_0\} = sY_0 - y_0(0^-) = sY_0$, because the system starts at rest. Multiplication by s is differentiation. Therefore, in the time domain,

$$y(t) = y_0(t) + \frac{1}{\alpha\zeta} \dot{y}_0(t)$$

where y_0 is the step response of the zero-free system.

With physical time and arbitrary ω_n , the coefficient is $1/(\alpha\zeta\omega_n) = 1/(\alpha\sigma)$. The displayed $1/(\alpha\zeta)$ form uses $\omega_n = 1$ (or differentiation with respect to normalised time). The zero initial value of y_0 is what permits $sY_0 = \mathcal{L}\{\dot{y}_0\}$ without an initial-condition term.



[FIG 3.30] — PDF p. 124 (slide: y , y_0 and the derivative term)

Say: "The derivative has a large hump early, while y_0 is still climbing. Adding that hump lifts the early response — which is exactly what more overshoot and a quicker rise look like."

[DEMO 3, continued] — the decomposition table

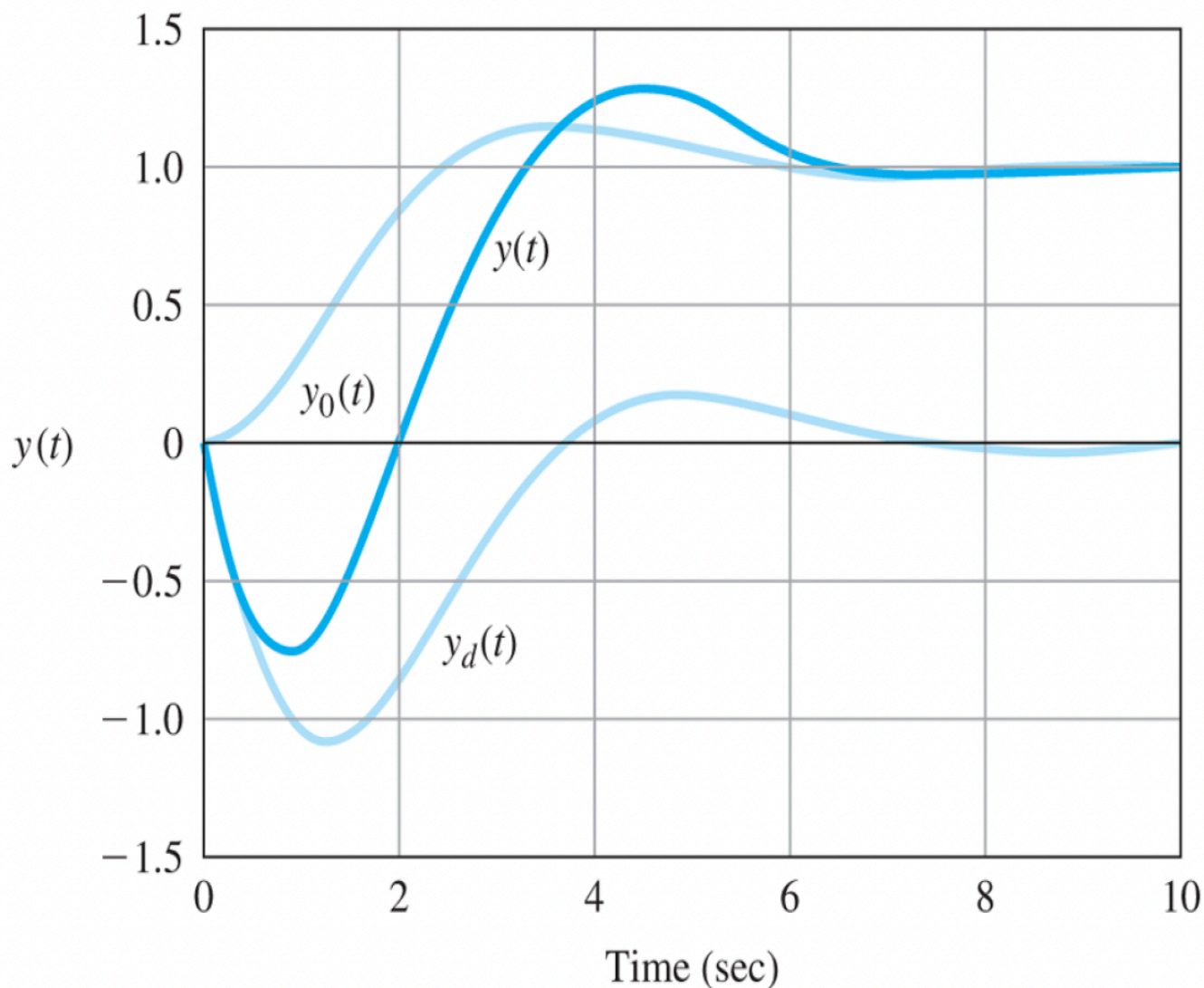
Teaching check [beyond the book]: Use the analytical derivative $\dot{y}_0 = h_0$ to check $y = y_0 + \dot{y}_0/(\alpha\zeta)$ when $\omega_n = 1$. This avoids mistaking finite-difference error for failure of the identity.

10. Right half-plane zeros

Now let $\alpha < 0$, so the zero sits at $s = +|\alpha|\sigma$. The numerator becomes $1 - s/(|\alpha|\zeta)$ and

$$y(t) = y_0(t) - \frac{1}{|\alpha|\zeta} \dot{y}_0(t)$$

The hump is subtracted. Early on, when $\dot{y}_0$ is largest, the response is pushed *down* — often below zero.



[FIG 3.31] — PDF p. 125 (slide)

One real RHP zero added to this standard pair $\Rightarrow$ the step response starts out in the wrong direction

This is **nonminimum-phase** behaviour. Here $\dot{y}(0^+) = 1/(\alpha\zeta) < 0$ in normalised units, which proves the initial reversal.

Derivation of the initial slope. Differentiate the decomposition: $\dot{y} = \dot{y}_0 + \frac{1}{\alpha\zeta}\dot{y}_d$. For the zero-free pair, $\dot{y}_0 = h_0$, which has $h_0(0^+) = 0$ by Eq. (3.66) because $\sin 0 = 0$. Its derivative at 0^+ is $\ddot{y}_0(0^+) = \omega_n^2 = 1$, from differentiating Eq. (3.66) or from the initial value theorem, $\lim_{s \rightarrow \infty} s^2 H_0(s) = 1$. So $\dot{y}(0^+) = 0 + \frac{1}{\alpha\zeta} \cdot 1 = 1/(\alpha\zeta)$, which is negative for $\alpha < 0$. In general, having RHP zeros does not always mean the step response's first motion is backwards; other zeros and the relative degree matter. For example, two real RHP zeros can give an initially positive response followed by an inverse excursion. The unstable zero-dynamics interpretation is developed in (0.0.8 §31–§34).

[DEMO 3, continued] — the RHP table

Teaching check [beyond the book]: For $\zeta = 0.5$, $\omega_n = 1$ and $\alpha = -1, -2, -4$, the minimum step-response values are approximately $-0.752, -0.280, -0.091$. The early negative excursion grows as the positive real zero approaches the origin.

10.1 A correction worth making [beyond the book]

The book's summary says a RHP zero "will depress the overshoot". Read carefully, because the comparison matters:

Comparison	Result
RHP zero at $\alpha = -2$ versus LHP zero at $\alpha = +2$	20.9% versus 29.8% — depressed

Comparison	Result
RHP zero at $\alpha = -2$ versus no zero at all	20.9% versus 16.3% — raised

For this family, against the same-distance LHP zero the statement holds. Against the zero-free system it does not: subtracting the derivative term *after* the peak, where $\dot{y}_0 < 0$, pushes the response up. The reliable lesson here is inverse response and a performance limitation, not a universal decrease in overshoot. Demo 3 prints all three numbers.

10.2 Example 3.28: the zero moving through the poles

$$H(s) = \frac{24}{z} \cdot \frac{s+z}{(s+4)(s+6)}, \quad z = 1, \dots, 6$$

The factor $24/z$ makes the DC gain unity: $H(0) = \frac{24}{z} \cdot \frac{z}{4 \cdot 6} = 1$. For a unit step,

$$Y(s) = \frac{24}{z} \cdot \frac{s+z}{s(s+4)(s+6)} = \frac{C_0}{s} + \frac{C_4}{s+4} + \frac{C_6}{s+6}.$$

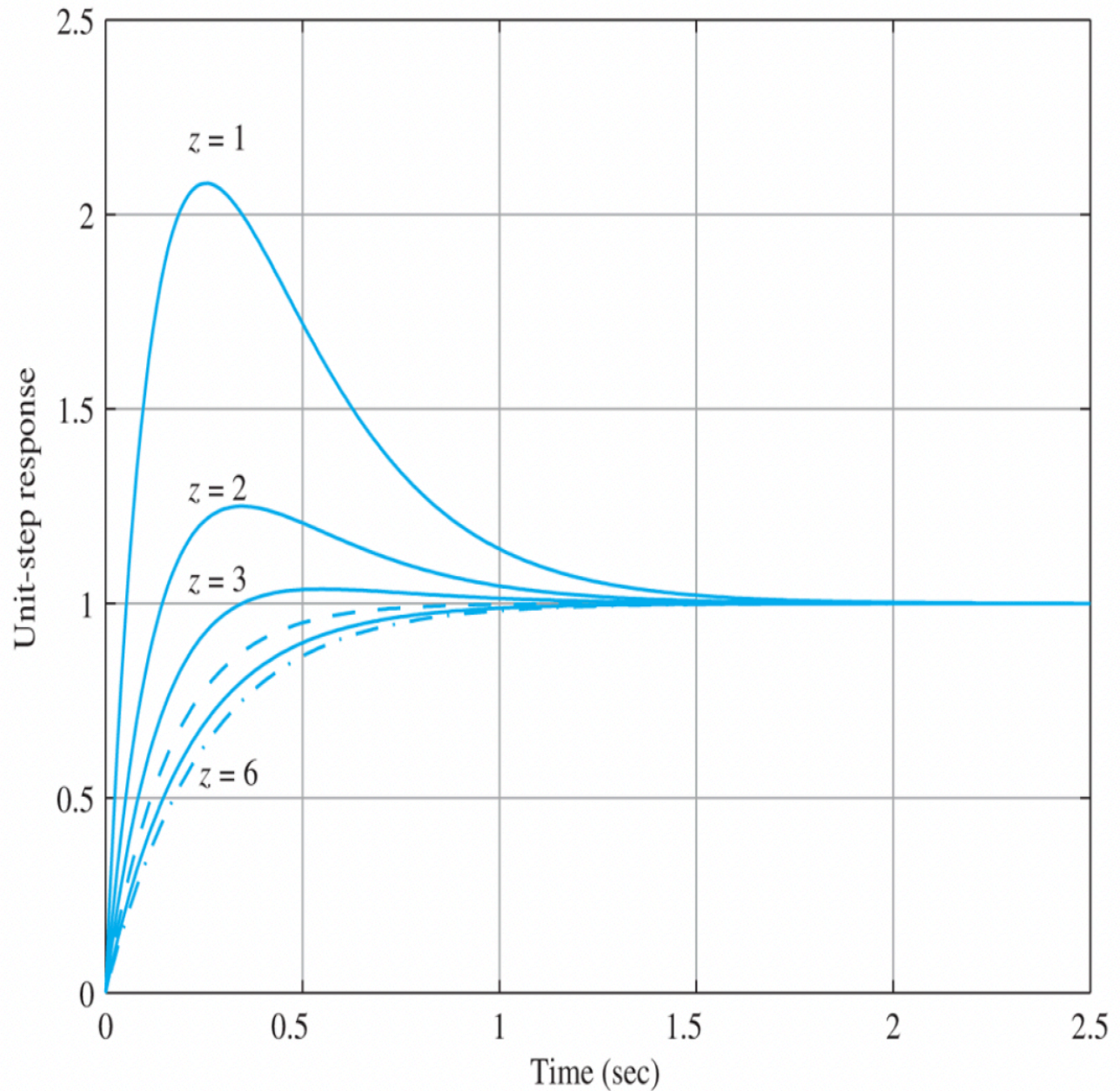
Cover up each factor:

$$C_0 = \frac{24}{z} \cdot \frac{z}{(4)(6)} = 1, \quad C_4 = \frac{24}{z} \cdot \frac{z-4}{(-4)(2)} = -\frac{3(z-4)}{z} = \frac{12}{z} - 3, \quad C_6 = \frac{24}{z} \cdot \frac{z-6}{(-6)(-2)} = \frac{2(z-6)}{z} = 2 - \frac{12}{z}.$$

So the step response is

$$y(t) = 1 + \left(\frac{12}{z} - 3\right)e^{-4t} + \left(2 - \frac{12}{z}\right)e^{-6t}.$$

Check: $y(0) = 1 + \frac{12}{z} - 3 + 2 - \frac{12}{z} = 0$ for every $z \checkmark$.



[FIG 3.32] — PDF p. 127 (slide)

Read the coefficients as z moves, from Demo 3's table:

z	coeff. of e^{-4t}	coeff. of e^{-6t}	overshoot
1	+9.000	−10.000	108%
2	+3.000	−4.000	25%
3	+1.000	−2.000	3.7%
4	0	−1.000	0
5	−0.600	−0.400	0
6	−1.000	0	0

The overshoot column, by hand [beyond the book]. Write c_4 and c_6 for the two coefficients. A peak needs $\dot{y} = -4c_4e^{-4t} - 6c_6e^{-6t} = 0$. Dividing by e^{-6t} gives $e^{2t_p} = -\frac{6c_6}{4c_4}$. This has a positive solution only when $c_4 > 0 > c_6$ and $-6c_6 > 4c_4$, which is the case $z < 4$. Let $r = e^{-2t_p} = -\frac{2c_4}{3c_6}$. Then $e^{-4t_p} = r^2$, $e^{-6t_p} = r^3$, and $y(t_p) = 1 + c_4r^2 + c_6r^3$:

z	c_4, c_6	r	$t_p = -\frac{1}{2} \ln r$	$y(t_p)$
1	9, -10	$\frac{3}{5}$	0.255 s	$1 + 9(0.36) - 10(0.216) = 2.08$
2	3, -4	$\frac{1}{2}$	0.347 s	$1 + \frac{3}{4} - \frac{4}{8} = 1.25$
3	1, -2	$\frac{1}{3}$	0.549 s	$1 + \frac{1}{9} - \frac{2}{27} = 1.037$

These reproduce 108%, 25% and 3.7%. For $4 \leq z \leq 6$ both coefficients are ≤ 0 , so $y < 1$ for all $t > 0$ and there is no overshoot.

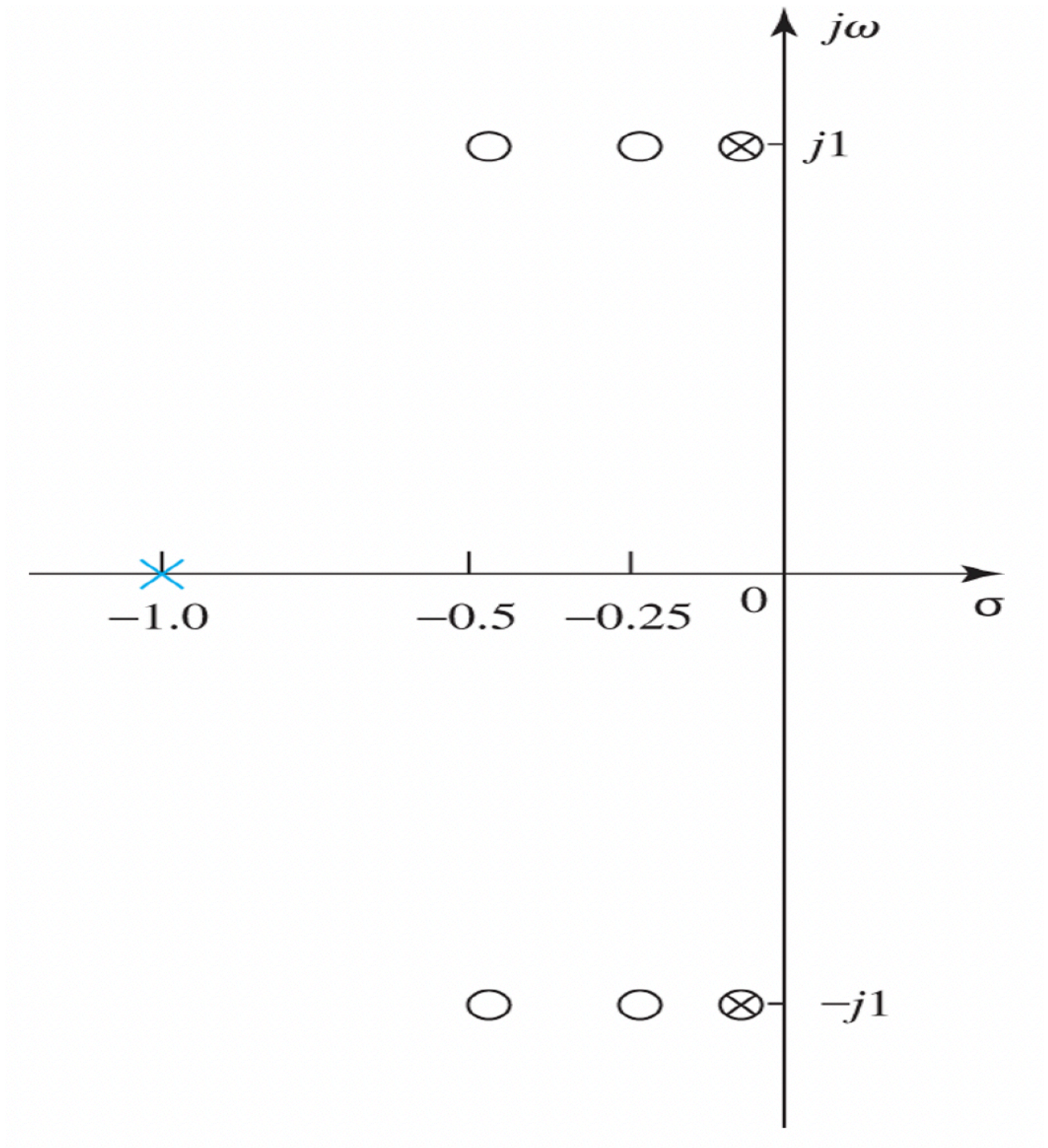
Three things to point at:

1. $z = 4$ **and** $z = 6$: a common factor cancels, its residue is *exactly* zero, and the input-output response is first order. If a physical second-order realisation retains that mode, it is hidden from this transfer function; the reduced transfer function alone does not establish its internal presence.
2. $z = 5$, **between the poles**: both coefficients are negative, the response approaches its final value from below, and there is no overshoot at all.
3. $z \leq 3$: the zero is nearer the origin than either pole, the coefficients grow and take opposite signs, and the overshoot runs away.

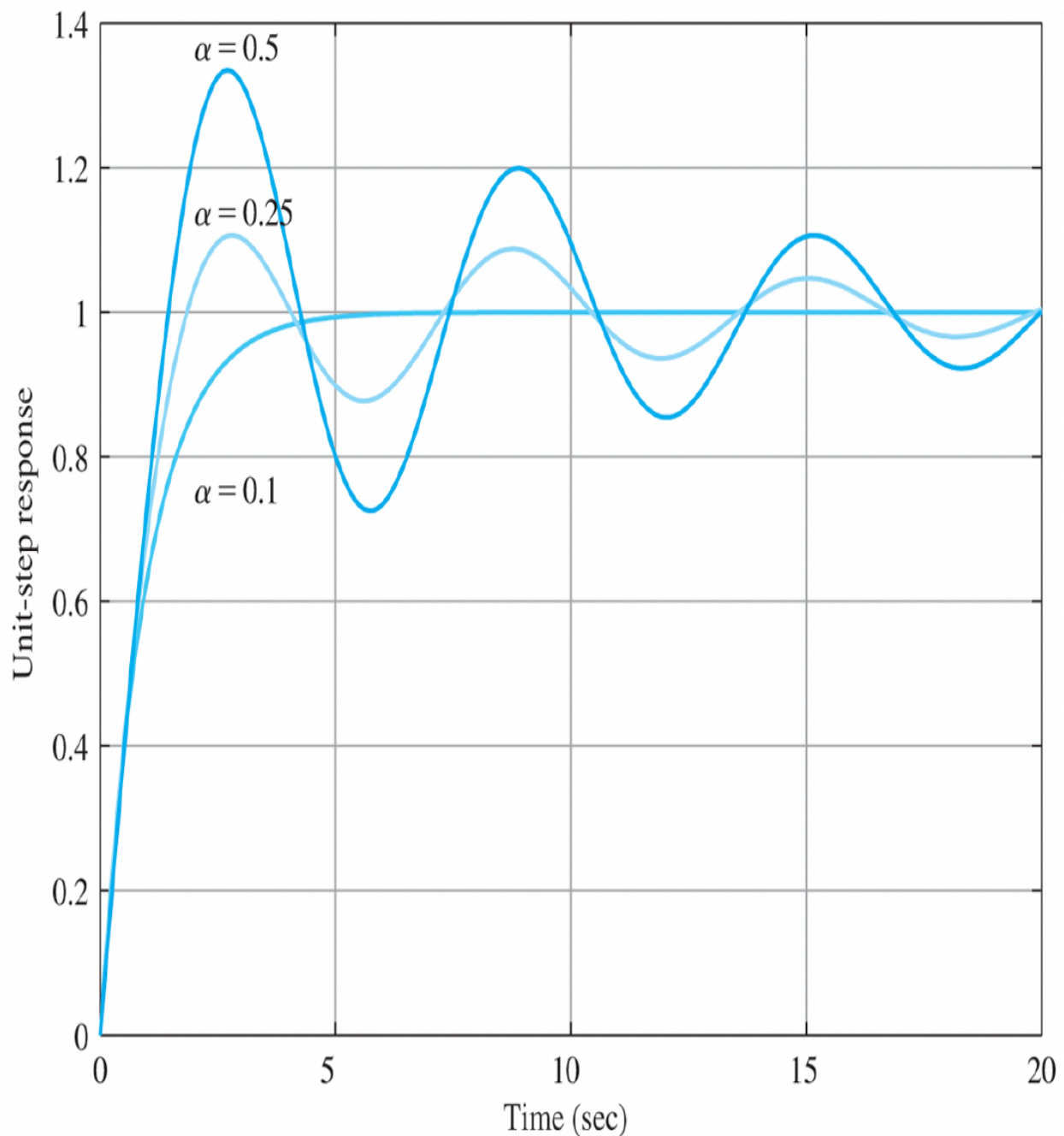
10.3 Example 3.29: complex zeros near lightly damped poles

$$H(s) = \frac{(s + \alpha)^2 + \beta^2}{(s + 1)[(s + 0.1)^2 + 1]}$$

with the poles at $-0.1 \pm j$ and the zeros placed at $-\alpha \pm j\beta$ for $(\alpha, \beta) = (0.1, 1.0)$, $(0.25, 1.0)$ and $(0.5, 1.0)$.



[FIG 3.33] — PDF p. 128 (the three zero locations)



[FIG 3.34] — PDF p. 130 (the three step responses)

Show: Fig. 3.33 first and ask the class to predict; then Fig. 3.34. **Point at:** the (0.1, 1.0) trace, where the zeros land exactly on the poles and the oscillation disappears completely, and the (0.5, 1.0) trace, where they do not and it does not.

Then read the book's warning, which is the whole lesson of the example:

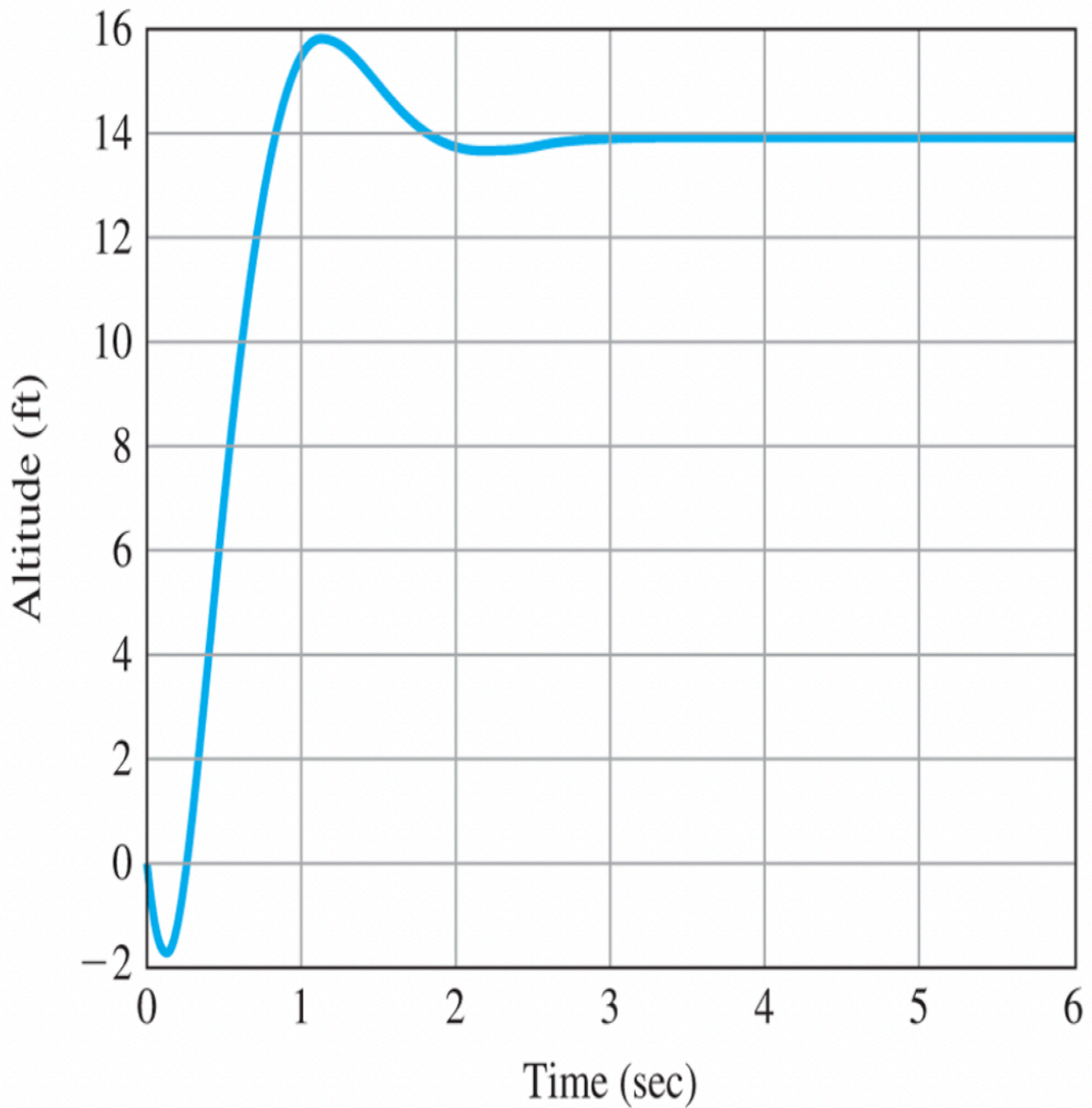
In practice, the locations of the lightly damped poles are not known precisely, and exact cancellation is not really possible.

[beyond the book] The stated transfer function has DC gain $H(0) = \frac{\alpha^2 + \beta^2}{1 \cdot (0.1^2 + 1)} = \frac{\alpha^2 + \beta^2}{1.01}$. For the three cases this is $1.01/1.01 = 1.00$, $1.0625/1.01 = 1.052$ and $1.25/1.01 = 1.238$. Fig. 3.34 is drawn about a common final value of 1, suggesting an unstated normalisation or a plotting inconsistency. To reproduce that comparison explicitly, plot $H(s)/H(0)$ and label it normalised. The source does not state the normalisation. Placing compensator zeros near a resonance can attenuate its response, but exact cancellation is sensitive to modelling error. L4 treats unstable cancellations.

11. Example 3.30: an aeroplane that descends when you pull up

$$\frac{h(s)}{\delta_e(s)} = \frac{30(s-6)}{s(s^2+4s+13)}$$

Altitude h from elevator angle δ_e , for a Boeing 747. A zero at $s = +6$, from $s - 6 = 0$. Poles at $s = 0$ and at the roots of $s^2 + 4s + 13$, which are $s = \frac{-4 \pm \sqrt{16-52}}{2} = -2 \pm 3j$.



[FIG 3.35] — PDF p. 132 (slide)

11.1 The physics, which is the point

Instructor script

By convention a negative elevator deflection is upward. Deflect the elevators up and you push the tail *down*, which rotates the nose up.

But rotation takes time. In the first instant, all you have done is add a downward force at the tail. The aeroplane sinks.

Then the nose comes up, the wings meet the air at a larger angle of attack, lift increases, and the aircraft climbs to a new altitude.

Down, then up. That is the right half-plane zero, and notice what it is not: it is not a modelling error, not a delay, and not something a better controller removes. It is the physics of where the control surface sits relative to the centre of mass. Compare the flexible-structure and quadrotor examples of the earlier lecture (0.0.8 §28–§30): the same geometry, the same conclusion.

11.2 The numbers

Final value, for the negative unit impulse $\delta_e(t) = -\delta(t)$, whose transform is $\Delta_e(s) = -1$:

$$h(\infty) = \lim_{s \rightarrow 0} s \cdot \frac{30(s-6)(-1)}{s(s^2+4s+13)} = \frac{30 \times (-6) \times (-1)}{13} = +13.8.$$

The s in front cancels the integrator pole, leaving $\frac{-30(s-6)}{s^2+4s+13}$ evaluated at $s=0$: $\frac{-30(-6)}{13} = \frac{180}{13} = 13.85$.

The initial dip, from the transform [beyond the book]. The altitude transform is $H_{\text{alt}}(s) = \frac{-30(s-6)}{s(s^2+4s+13)}$, which has relative degree 2. The initial value theorem gives $h(0^+) = \lim_{s \rightarrow \infty} sH_{\text{alt}} = 0$ and $\dot{h}(0^+) = \lim_{s \rightarrow \infty} s^2 H_{\text{alt}} = -30$. The aircraft starts at the reference altitude and moves **down** at first, as the physics said.

Note why an impulse gives a finite final value: the pole at the origin integrates, and the integral of an impulse is a constant. A step input would give an altitude that increases forever.

Here $sH_{\text{altitude}}(s)$ has poles $-2 \pm 3j$, so the Final Value Theorem applies to this impulse response even though the altitude plant itself has an integrator and is not BIBO stable. After scaling, the altitude impulse response equals the step response of a second-order pair with a RHP zero (see Demo 4 below). Match $s^2 + 4s + 13$ to $s^2 + 2\zeta\omega_n s + \omega_n^2$. This gives $\omega_n = \sqrt{13} = 3.61$ rad/s, $2\zeta\omega_n = 4$ so $\zeta = 2/\sqrt{13} = 0.555$, and $\sigma = \zeta\omega_n = 2$, agreeing with the real part of the roots.

The estimates in the table follow:

- $t_r \simeq 1.8/3.606 = 0.50$ s.
- For overshoot, $\sqrt{1-\zeta^2} = \sqrt{9/13} = 3/\sqrt{13}$, so the exponent is $\pi\zeta/\sqrt{1-\zeta^2} = \pi \cdot \frac{2}{3} = 2.094$ and $M_p = e^{-2.094} = 0.123$.
- $t_s \simeq 4.6/2 = 2.30$ s.

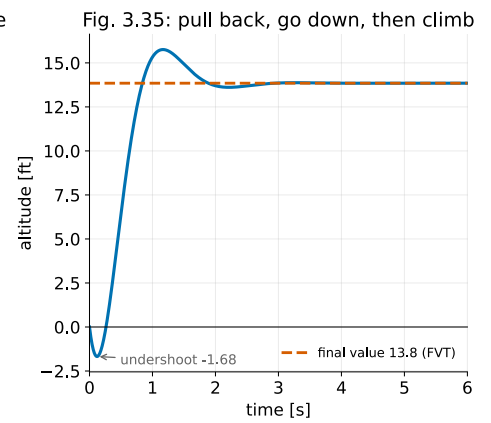
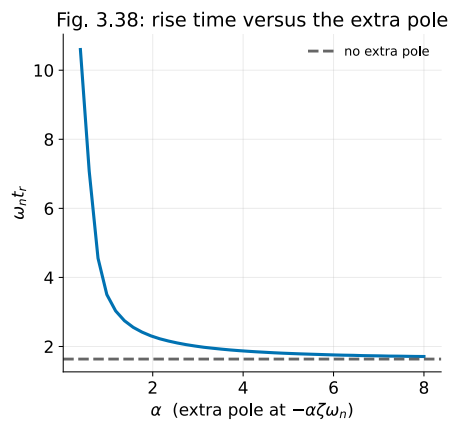
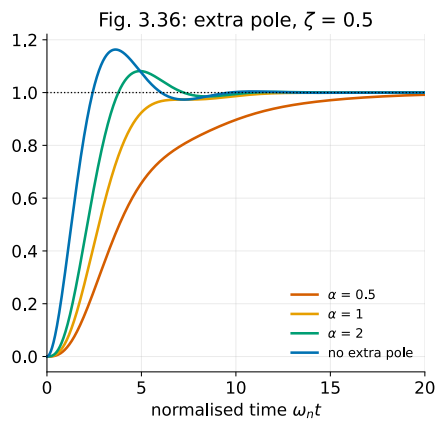
Quantity	Estimate	Source	Recomputed from the model (Demo 4)
t_r	0.50 s	$1.8/\omega_n$	0.43 s
M_p	12.3%	Eq. (3.72), using $\zeta = 2/\sqrt{13}$	13.8%
t_s	2.30 s	$4.6/\sigma$	2.54 s

Attribution [beyond the book]: PDF p. 133 gives a coarse zero-free overshoot estimate of 14% from Fig. 3.24; the table instead evaluates Eq. (3.72), giving 12.3%. The book reports the aircraft response as approximately 0.43 s, 14%, and 2.6 s; our recomputation gives 0.429 s, 13.805%, and 2.535 s, rounded in the last column.

[DEMO 4] – [ch3/l3_demo4_extra_pole_aircraft.py](#) (live, ~10 s)

Teaching check [beyond the book]: For the negative unit elevator impulse, verify $h(\infty) = 180/13 \approx 13.846$ and an initial minimum near -1.68 . The altitude transform is $\frac{180}{13} \frac{1-s/6}{s} \frac{13}{s^2+4s+13}$. To see this, factor $-30(s-6) = 180(1-s/6)$ and split $180 = \frac{180}{13} \cdot 13$. After scaling, this impulse response is exactly the step response of the second-order pair with a RHP zero: $1/s$ is the step, and the rest is a unit-DC-gain pair with the zero at $+6$. This explains why the second-order estimates are useful here.

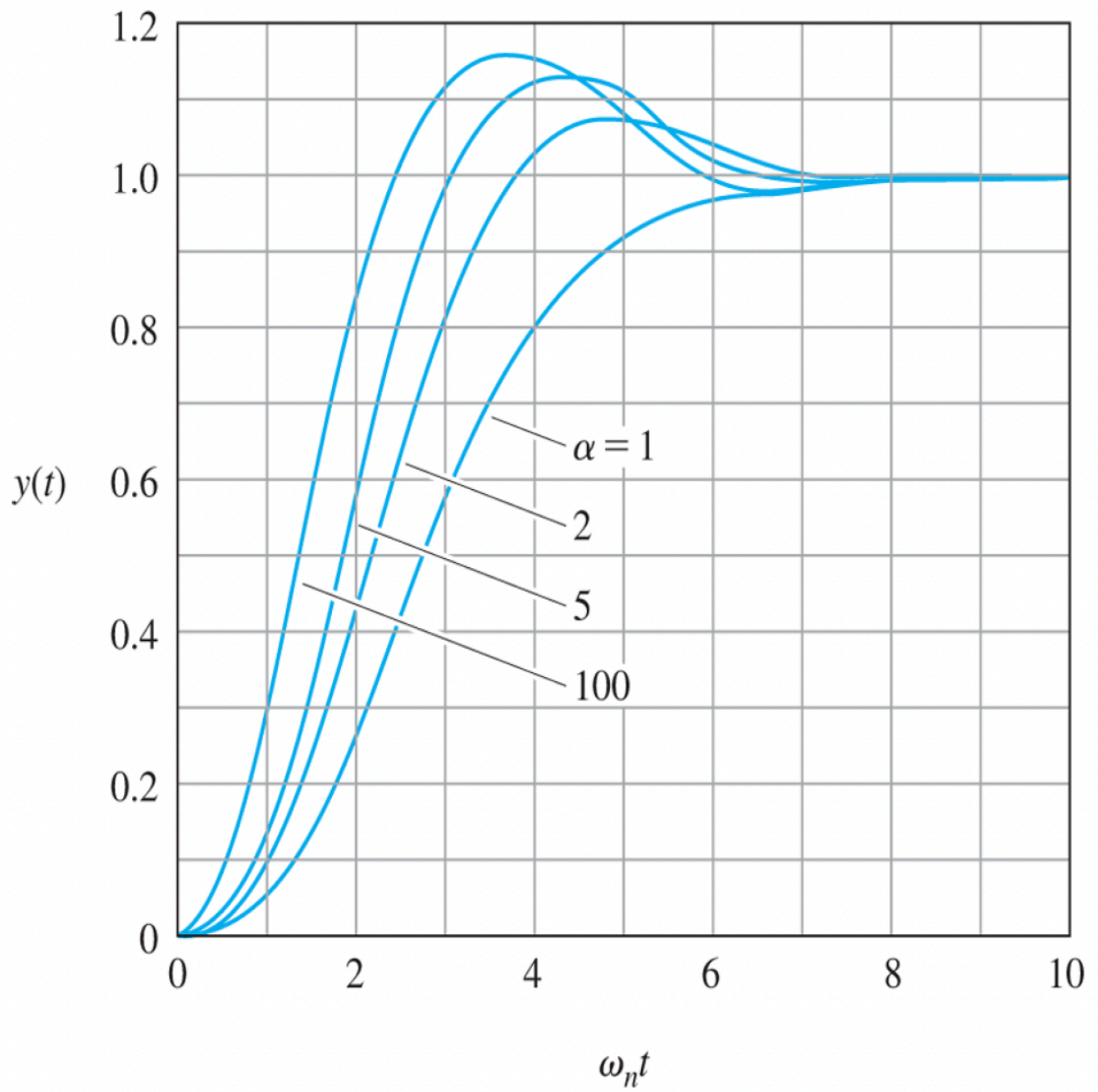
Beyond the pure second order: an extra pole and a RHP zero

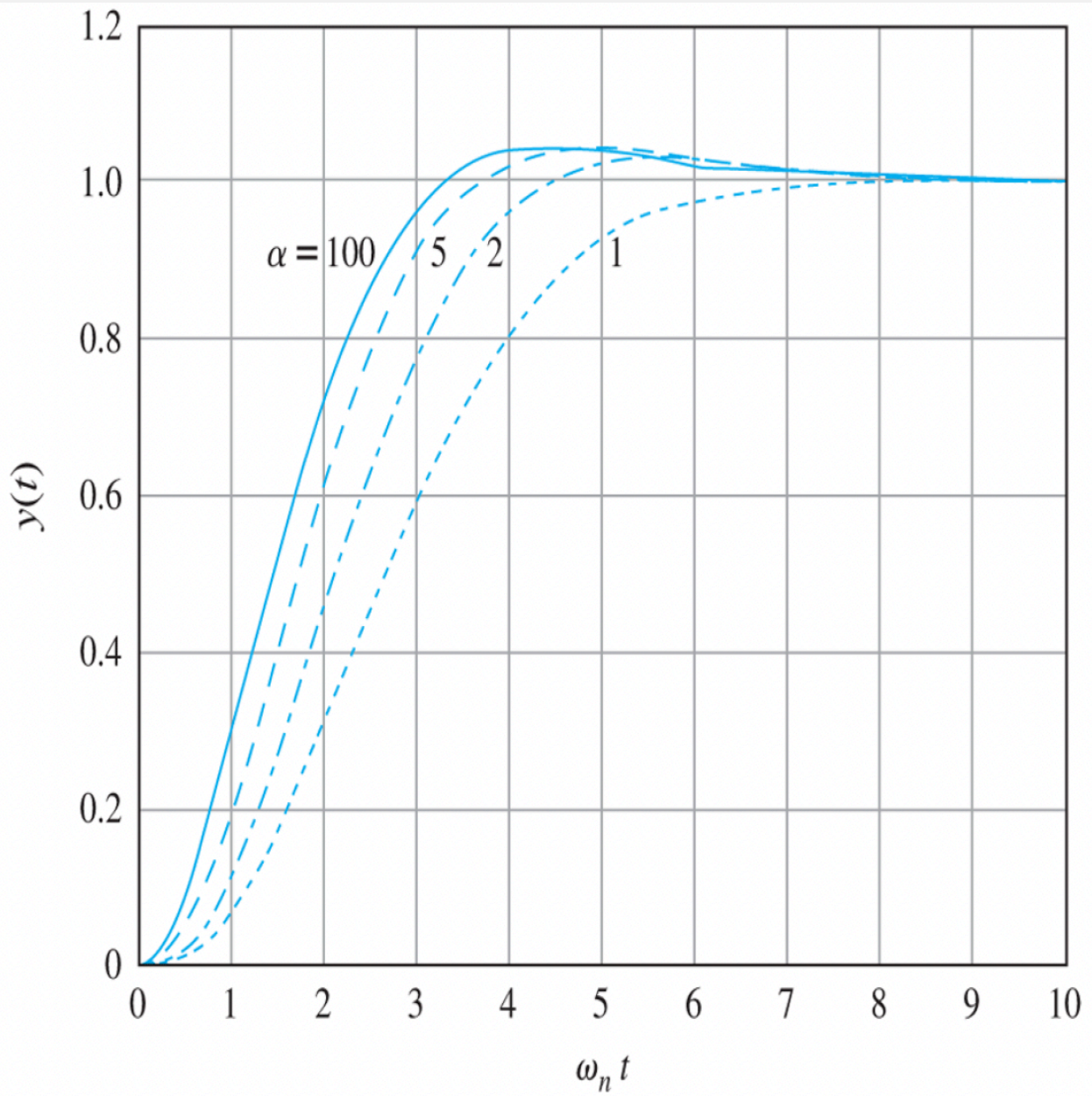


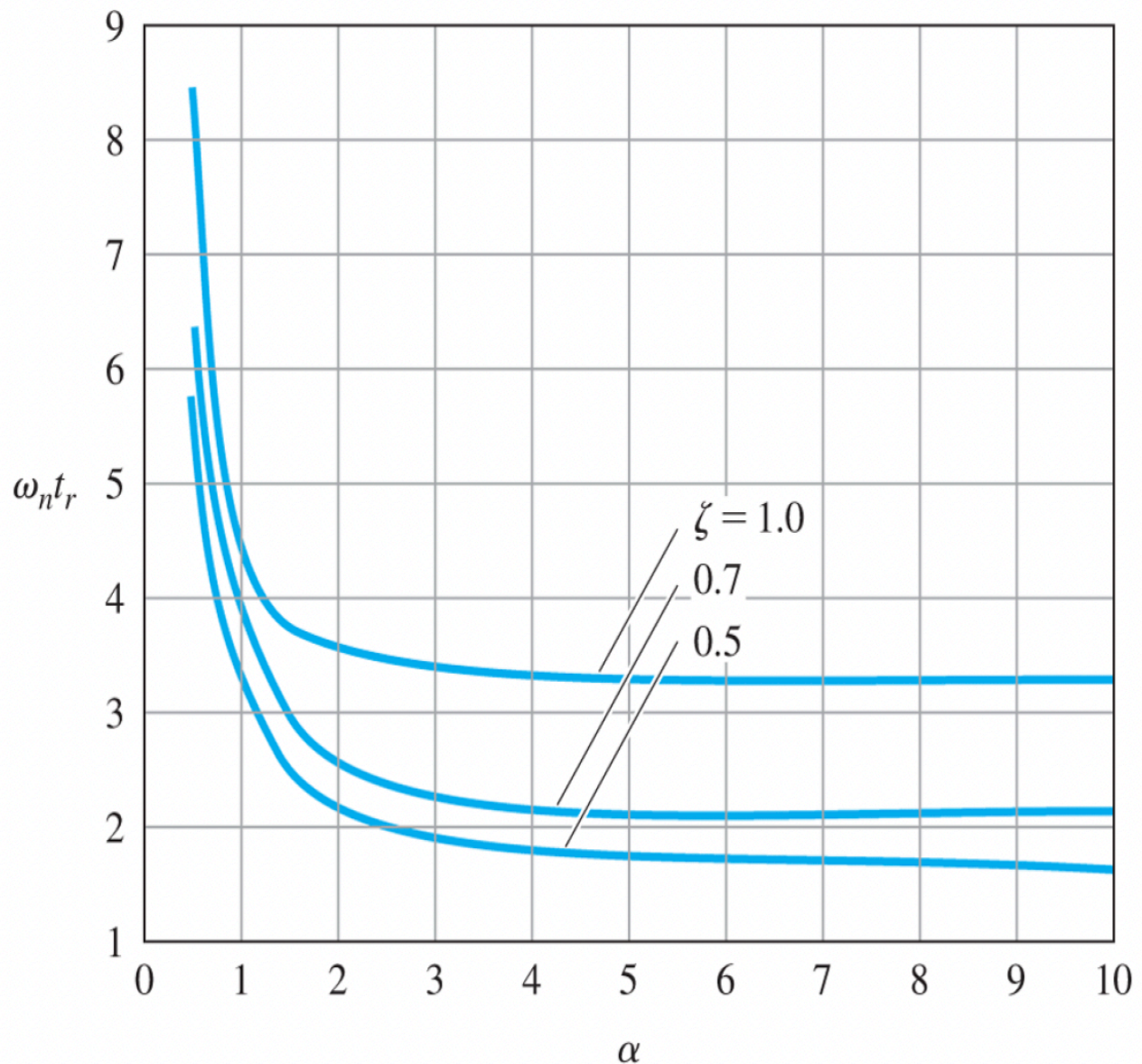
12. Extra poles

Add a real pole at $-\alpha\zeta\omega_n$ to the standard pair:

$$H(s) = \frac{1}{\left(\frac{s}{\alpha\zeta\omega_n} + 1\right) \left[\left(\frac{s}{\omega_n}\right)^2 + 2\zeta\frac{s}{\omega_n} + 1\right]} \quad (3.82)$$



[FIG 3.36] — PDF p. 134 ($\zeta = 0.5$)[FIG 3.37] — PDF p. 135 ($\zeta = 0.707$)



[FIG 3.38] — PDF p. 136 (normalised rise time versus α)

an extra LHP pole increases the rise time; a LHP zero decreases it

[DEMO 4, continued] — left and middle panels

Teaching check [beyond the book]: For $\zeta = 0.5$, the normalised 10–90% rise times with an added pole at $-\alpha\sigma$ are approximately 1.87, 2.29, 3.46, and 8.49 for $\alpha = 4, 2, 1, 0.5$, compared with 1.64 without the extra pole. Even $\alpha = 4$ changes rise time by about 14%.

Use the same factor-of-four heuristic to identify extra poles worth checking, not to certify that more distant poles have no effect. Approximation accuracy depends on modal residues, zeros, and the required tolerance as well as pole separation.

12.1 By the way: sharper estimates exist [beyond the book]

Scope. The lecture stays with FPE's three formulas. They are deliberately simple, and their simplicity is what turns specifications into a radius, an angle and a distance (§6). This subsection is an optional aside and is not in the 75-minute plan. In lecture it deserves one sentence at most: "sharper estimates exist; they are in the notes." The student version is §9.1 of the student notes. It gives the results without derivations.

The asides below cover three refinements: a better rise-time fit, an exact bracket on settling time, and a residue correction that accounts for zeros and extra poles.

(a) Rise time: better curve fits

Other textbooks fit the normalised 10–90% rise time more closely than 1.8:

$$\omega_n t_r \simeq 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1 \quad (\text{Nise, 7th ed., §4.6, Fig. 4.16})$$

$$\omega_n t_r \simeq 2.16\zeta + 0.60, \quad 0.3 \leq \zeta \leq 0.8 \quad (\text{Dorf and Bishop, 13th ed., Eq. 5.17})$$

Nise obtained the cubic with MATLAB's `polyfit`. Both are empirical fits, not derivations. Compared with the measured values from §5.2:

ζ	measured $\omega_n t_r$	1.8	Nise cubic	Dorf linear
0.1	1.104	+63%	1.101 (−0.2%)	0.816 (−26%)
0.3	1.321	+36%	1.322 (0.0%)	1.248 (−5.5%)
0.5	1.638	+10%	1.635 (−0.1%)	1.680 (+2.6%)
0.7	2.126	−15%	2.127 (0.0%)	2.112 (−0.7%)
0.9	2.883	−38%	2.880 (−0.1%)	2.544 (−12%)
1.0	3.358	−46%	3.382 (+0.7%)	2.760 (−18%)

The price of accuracy. The bound $\omega_n \geq 1.8/t_r$ is a circle. With either fit, the rise-time boundary becomes $\omega_n \geq f(\zeta)/t_r$, whose radius grows with ζ . The shaded region of Example 3.27 is then no longer a disc removed from a wedge. The rise-time failure of Demo 2 (§6.1) is exactly the error this fit removes.

Ogata (5th ed., Eq. 5-19) derives an exact rise time, $t_r = (\pi - \cos^{-1} \zeta)/\omega_d$. It uses the 0–100% definition, however, so it is not comparable with FPE's 10–90% convention.

(b) Settling time: no smooth formula can be exact, but the exact value can be bracketed

Why no smooth formula works. Ogata (5th ed., Fig. 5-11) plots the exact settling time against ζ and remarks that it is discontinuous. As ζ increases, an oscillation peak can drop inside the band. When that happens, the last exit from the band jumps back by about half a period. For the 1% band, the measured σt_s jumps between about 3.8 and 5.1 over $0.3 \leq \zeta \leq 0.9$. The book's 4.6 tracks the middle of this staircase.

The exact bracket. Assume the standard pair of Eq. (3.69), with $0 < \zeta < 1$, unit final value and a band $0 < \epsilon < 1$.

Step 1 (extrema). From §5.3, $\dot{y} = \frac{\omega_n^2}{\omega_d} e^{-\sigma t} \sin \omega_d t$. The extrema therefore occur at $\omega_d t_k = k\pi$ for $k = 1, 2, \dots$. Substituting into Eq. (3.69), $\sin k\pi = 0$ and $\cos k\pi = (-1)^k$, so

$$y(t_k) - 1 = -(-1)^k e^{-\sigma k\pi/\omega_d} = -(-1)^k M_p^k,$$

using $e^{-\sigma\pi/\omega_d} = M_p$ from §5.3. Take $t_0 = 0$ as well, where $y - 1 = -1 = -M_p^0$.

Step 2 (monotone between extrema). On (t_k, t_{k+1}) , $\sin \omega_d t$ has one sign, so y is monotone there. By Eq. (3.70), $y - 1$ vanishes where $\cos(\omega_d t - \beta) = 0$. Since $0 < \beta < \pi/2$, exactly one such point lies in the interval, at $\omega_d t = k\pi + \pi/2 + \beta$. So $|y - 1|$ falls monotonically from M_p^k to zero, then rises monotonically to M_p^{k+1} .

Step 3 (the bracket). Let k^* be the largest integer $k \geq 0$ with $M_p^k > \epsilon$. Equivalently, $k^* = \lceil \ln \epsilon / \ln M_p \rceil - 1$.

- At t_{k^*} , $|y - 1| = M_p^{k^*} > \epsilon$, so the response is still outside the band and $t_s > t_{k^*}$.
- After the zero crossing that follows t_{k^*} , Step 2 gives $|y - 1| \leq M_p^{k^*+1} \leq \epsilon$. This holds on the rest of that interval and on every later one. The last exit from the band therefore occurs before that zero crossing.

Hence

$$k^*\pi < \omega_d t_s < k^*\pi + \frac{\pi}{2} + \sin^{-1} \zeta, \quad k^* = \left\lceil \frac{\ln \epsilon}{\ln M_p} \right\rceil - 1$$

Worked check. Take $\zeta = 0.5$ and $\epsilon = 0.01$.

- $M_p = 0.163$ and $\ln 0.01 / \ln 0.163 = -4.605 / -1.814 = 2.54$, so $k^* = 2$.
- $\sigma/\omega_d = 0.5/0.866 = 0.577$.

- The lower end is $\sigma t_s > 0.577 \times 2\pi = 3.628$.
- The upper end is $\sigma t_s < 0.577 \times (2\pi + \pi/2 + \pi/6) = 0.577 \times 8.378 = 4.837$.
- The measured value is $\sigma t_s = 4.390$, inside the bracket.

A SciPy sweep confirmed the bracket at every ζ from 0.30 to 0.90 in steps of 0.05; for example:

ζ	measured σt_s	k^*	bracket on σt_s
0.45	3.950	2	(3.166, 4.193)
0.50	4.390	2	(3.628, 4.837)
0.60	3.755	1	(2.356, 4.017)
0.70	4.602	1	(3.079, 5.379)
0.85	3.793	0	(0, 4.174)

When ζ goes from 0.55 to 0.60, k^* drops from 2 to 1, and the measured value falls from 4.73 to 3.76. That drop is the staircase described above.

(c) Zeros and extra poles: the dominant-residue correction

This is the most useful of the three refinements. It extends Eqs. (3.71)–(3.72) to a dominant pair with extra zeros and poles, using only the cover-up idea of §7.

Assumptions.

- The complete model is $H(s) = H_0(s)G(s)$, where $H_0 = \omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$ is the standard pair with $0 < \zeta < 1$.
- $G(0) = 1$, so the DC gain is 1.
- The poles are simple, and no pole or zero of G sits at the origin.
- The zeros z_i and extra poles p_j of G are written as locations in the s -plane:

$$G(s) = \prod_i \frac{s - z_i}{-z_i} \prod_j \frac{-p_j}{s - p_j}.$$

Let $p = -\sigma + j\omega_d$ be the upper pole of the pair and $\bar{p}$ its conjugate.

Step 1 (cover-up). The unit-step response is $Y = H/s$. Its residue at p is

$$R = \left. \frac{\omega_n^2 G(s)}{s(s - \bar{p})} \right|_{s=p} = R_0 G(p), \quad R_0 = \left. \frac{\omega_n^2}{s(s - \bar{p})} \right|_{s=p},$$

where R_0 is the residue for H_0 alone. **The extra zeros and poles multiply the pair's residue by the single complex number $G(p)$.**

Step 2 (the pair's term). For H_0 alone, the pair contributes $2 \operatorname{Re}(R_0 e^{pt}) = y_0 - 1 = -e^{-\sigma t} \cos(\omega_d t - \beta) / \sqrt{1 - \zeta^2}$ by Eq. (3.70). Write $G(p) = |G(p)|e^{j\phi}$. Since $e^{pt} = e^{-\sigma t} e^{j\omega_d t}$, the factor $e^{j\phi}$ adds ϕ to $\omega_d t$ and leaves $e^{-\sigma t}$ unchanged:

$$2 \operatorname{Re}(R_0 |G(p)| e^{j\phi} e^{pt}) = -|G(p)| \frac{e^{-\sigma t}}{\sqrt{1 - \zeta^2}} \cos(\omega_d t + \phi - \beta).$$

The complete response is therefore

$$y(t) = 1 - |G(p)| \frac{e^{-\sigma t}}{\sqrt{1 - \zeta^2}} \cos(\omega_d t + \phi - \beta) + \sum_j C_j e^{p_j t},$$

where the last sum collects the modes of the extra poles.

Step 3 (the approximation). Drop $\sum_j C_j e^{p_j t}$ near the peak. **If G has zeros only, nothing is dropped and the result is exact.**

Let $\theta = \omega_d t + \phi$. Differentiating the remaining term, and using $\sigma = \omega_n \sin \beta$ and $\omega_d = \omega_n \cos \beta$,

$$\frac{d}{dt} \left[e^{-\sigma t} \cos(\theta - \beta) \right] = -e^{-\sigma t} [\sigma \cos(\theta - \beta) + \omega_d \sin(\theta - \beta)] = -\omega_n e^{-\sigma t} \sin \theta.$$

The extrema are at $\theta = k\pi$, that is, $\omega_d t_k = k\pi - \phi$. There $\cos(k\pi - \beta) = (-1)^k \cos \beta = (-1)^k \sqrt{1 - \zeta^2}$, so

$$y(t_k) - 1 = (-1)^{k+1} |G(p)| e^{-\sigma t_k}.$$

Odd k gives overshoots and even k gives undershoots. The peak time is the first odd k with $t_k > 0$. When $\pi - \phi > 0$ this is $k = 1$, which covers every case below.

Step 4 (geometry). Take the argument of each factor of $G(p)$. For a real LHP root r , $-r > 0$ contributes no angle. Let $\psi_r = \arg(p - r) \in (0, \pi)$ be the angle, measured from the positive real axis, of the vector from the root to p . Then $\phi = \sum \psi_{\text{zeros}} - \sum \psi_{\text{poles}}$. A real RHP zero also contributes $-\pi$, because $-z_i < 0$. The magnitudes are distances: ℓ from the root to p , and d from the root to the origin.

$$t_p \simeq \frac{\pi - \sum \psi_{\text{zeros}} + \sum \psi_{\text{poles}}}{\omega_d}, \quad M_p \simeq \left[\prod_{\text{zeros}} \frac{\ell_i}{d_i} \prod_{\text{poles}} \frac{d_j}{\ell_j} \right] e^{-\sigma t_p}$$

With no extra roots, $G = 1$ and $\phi = 0$, and the box reduces to Eqs. (3.71)–(3.72). For a complex pair of zeros or poles, include both members, each with its own angle and distances.

Reading the result.

- A LHP zero adds angle, so the peak comes earlier, while the envelope $e^{-\sigma t}$ is still large. It also rescales the amplitude by ℓ/d . At $\alpha = 4$ that factor is 0.866, yet overshoot still rises from 16.3% to 19.1% because the peak moves earlier.
- An extra pole delays the peak and multiplies by d/ℓ .
- The angle sums preview the root-locus angle condition of Chapter 5.

Worked checks ($\zeta = 0.5$, $\omega_n = 1$, so $\sigma = 0.5$ and $\omega_d = 0.866$):

- *LHP zero, $\alpha = 2$ (zero at -1).* $p - z = 0.5 + 0.866j$, so $\ell = 1$, $d = 1$ and $\psi = 60^\circ$. Then $t_p = (180^\circ - 60^\circ)/\omega_d = 2.094/0.866 = 2.418$ and $M_p = e^{-0.5 \times 2.418} = e^{-1.209} = 0.298$. This is the 29.8% of §8, and it is exact.
- *RHP zero, $\alpha = -2$ (zero at $+1$).* $p - z = -1.5 + 0.866j$, so $\ell = \sqrt{3}$, $d = 1$, $\psi = 150^\circ$ and $\phi = 150^\circ - 180^\circ = -30^\circ$.
 - Overshoot ($k = 1$): $t_p = (180^\circ + 30^\circ)/\omega_d = 3.665/0.866 = 4.232$ and $M_p = \sqrt{3} e^{-2.116} = 0.209$. This is the 20.9% of §10.1.
 - Undershoot ($k = 0$): $t_0 = (30^\circ)/\omega_d = 0.524/0.866 = 0.605$, and $y - 1 = -\sqrt{3} e^{-0.302} = -1.280$, so $y_{\min} = -0.280$. This is the minimum quoted in §10.
- *Extra pole.*
 - $\alpha = 4$ (pole at -2): $p - p_j = 1.5 + 0.866j$, so $\ell = \sqrt{3}$, $d = 2$ and $\psi = 30^\circ$. Then $t_p = (180^\circ + 30^\circ)/\omega_d = 4.232$ and $M_p = (2/\sqrt{3}) e^{-2.116} = 0.139$. The simulation also gives 13.9%.
 - $\alpha = 2$: the estimate is 8.9% against 8.1% measured. The pole's own mode is no longer negligible at t_p .
 - $\alpha = 1$: the estimate is 3.8% against no overshoot at all. The real pole now dominates, and the approximation fails, as the factor-of-four rule warns.
- *The 747 (Ex. 3.30).* By the §11.2 worked check, the altitude response is the step response of the unit-DC pair $-2 \pm 3j$ with a zero at $+6$.
 - Geometry: $p - z = -8 + 3j$, so $\ell = \sqrt{73} = 8.544$, $d = 6$, $|G| = 1.424$, $\psi = 159.44^\circ$ and $\phi = -20.56^\circ = -0.3588$ rad.
 - Overshoot: $t_p = (\pi + 0.3588)/3 = 1.167$ s and $M_p = 1.424 e^{-2 \times 1.167} = 0.138$. This is exactly the complete-model 13.8%.
 - Undershoot: $t_0 = 0.3588/3 = 0.120$ s and $y - 1 = -1.424 e^{-0.239} = -1.121$. Scaled by 180/13, the minimum is -1.68 .
 - The whole gap between the pair-only 12.3% and the complete-model 13.8% in the §11.2 table is the RHP zero's residue factor.

Verification. SymPy confirmed three identities: the residue identity $R = R_0 G(p)$, the phase-shift form in Step 2, and the extremum values $(-1)^{k+1} |G| e^{-\sigma t_k}$. SciPy step responses reproduced every number above to the precision shown.

13. The summary box and closing

The book gathers §3.5 into four statements. Use this qualified summary so it agrees with the worked examples:

Source precision note [beyond the book]: the summary on PDF p. 137 gives approximately 35% overshoot at $\zeta = 0.3$; Eq. (3.72) gives 37.23%, rounded to 37% below.

1. Second order, no zeros: $t_r \simeq 1.8/\omega_n$, $M_p \approx 5\%, 16\%, 37\%$ at $\zeta = 0.7, 0.5, 0.3$, $t_s \simeq 4.6/\sigma$
2. A nearby real LHP zero increases overshoot in the standard family
3. A real RHP zero in that family causes inverse response; compare peaks explicitly
4. An extra real LHP pole slows the rise; factor 4 is a heuristic, not a bound

Closing script

We turned three customer requirements into three numbers: a radius, an angle, and a distance from the imaginary axis. Their intersection is a region of the s -plane, and every design method in the rest of this course is a way of moving poles into that region.

Then we checked the limits of that region. Nearby zeros change modal amplitudes, and an extra pole can slow the rise. The single real RHP zero in our example sends the response backwards first. Cancelling a plant's unstable zero with an unstable controller pole is not a route to internal stability.

Notice what none of this changed: the poles. A zero reweights the modes; it does not move them. That is why we can design with poles and then check with a simulation, rather than the other way around.

One question is still open, and it is the one that makes control engineering harder than it looks. Everything today assumed the response settles at all. Next time: how do we know it does, and how much gain can we use before it does not?

Part II — Materials

14. One-board summary

```
SECOND ORDER, NO ZEROS          (this is the reference case, and only this one)

t_r ~ 1.8 / wn                  <- a FIT; exact value of wn*t_r is 1.64 at zeta=0.5,
                                2.13 at zeta=0.7.  Calibrated near zeta = 0.6.
t_p = pi / omega_d              <- exact
M_p = exp(-pi zeta / sqrt(1 - zeta^2))  <- exact.  16% at 0.5, 5% at 0.7
t_s ~ 4.6 / sigma               <- approximate 1% rule, NOT a guaranteed bound
envelope bound = -ln(0.01*sqrt(1-zeta^2))/sigma,  0 < zeta < 1

SPECIFICATIONS -> REGION

t_r <= T   -> wn   >= 1.8/T      outside a circle
M_p <= M   -> zeta >= zeta(M)    inside a wedge, theta = asin(zeta)
t_s <= S   -> sigma >= 4.6/S    left of a vertical line

... then SIMULATE.  The circle is a fit; a boundary pole pair can fail.

ZEROS AND EXTRA POLES          zero at -alpha*sigma

y(t) = y0(t) + (1/(alpha sigma)) * y0'(t)      <- physical time
for wn = 1: sigma = zeta, giving the normalised formula

alpha > 0 (LHP zero) : add the derivative hump  -> more overshoot, faster rise
alpha < 0 (RHP zero) : subtract it               -> starts BACKWARDS, sluggish
extra LHP pole       -> slower rise

factor-of-4 rule: a screening heuristic; check error against the specification
exact cancellation: no contribution in this transfer; inspect internal modes
```

15. Discussion questions

1. M_p depends on ζ alone while t_r and t_s depend on ω_n and σ . What does that say about which specification you should negotiate with a customer first?
2. A pole pair sits exactly on the $\omega_n = 1.8/t_r$ circle at $\zeta = 0.9$ and fails the rise-time requirement. Has the design method failed?
3. Example 3.28 has the zero exactly on a pole for $z = 4$. If you built that system and pushed on it, would the missing mode be gone?
4. Two plants have identical poles. One has a zero at -1 , the other at $+1$. Which is harder to control, and what in the step response tells you so?
5. The 747 climbs to a finite altitude after an impulsive elevator input but would climb forever after a step. Which pole is responsible, and what does that mean for the pilot?
6. When would you deliberately place a compensator zero near a lightly damped pole, given Example 3.29's warning?

16. Homework and follow-up problems

Problem	Topic	Why this one
3.16	DC gain and final value of a second-order system	Short review before using unit-gain response formulas.
3.30, 3.31	Feedback parameters and attainable pole regions	Tests which specifications can be met simultaneously.
3.25, 3.26	Choosing gain and pole location for a unity-feedback system	First taste of design in the region.
3.27	Peak time requirement	Uses Eq. (3.71) alone; short and clean.
3.28, 3.29	Dynamics dominated by a complex pair; multiple specifications	The Example 3.27 workflow with new numbers.
3.36	Initial-condition response and logarithmic decrement	Relates observed decay to damping ratio.
3.37	Ideal pitch response, aircraft	The Example 3.30 setting as a design problem.
3.38	Approximating higher-order systems by second-order ones	The judgement §12 asks for.
3.39, 3.42	Modal response form, settling time, and overshoot estimates	State assumptions before applying the formulas.
3.41	Sketch a step response, then compare with Matlab	Predict from poles and zeros before computing.

Suggested additional exercise [beyond the book]: compare the altitude impulse responses of $-30(s-6)/[s(s^2+4s+13)]$ and $30(s+6)/[s(s^2+4s+13)]$. They have the same poles and final altitude $180/13$, so the comparison isolates zero location. Report undershoot, rise time, and overshoot.

Answer key (numerical, 1% settling).

Zero	Undershoot	t_r	M_p	t_s
+6 (RHP, the 747)	-1.68	0.43 s	13.8%	2.54 s
-6 (LHP mirror)	none	0.38 s	15.8%	2.24 s

Both zeros are $6 = 3\sigma$ from the imaginary axis ($|\alpha| = 3$ in the normalised family), so both overshoot a little more than the zero-free 12.3%. Only the RHP zero dips first and rises more slowly.

17. Instructor cautions

1. **State the settling-time convention and approximation.** The decay-rate estimates are $4.6/\sigma$, $3.9/\sigma$, and $3.0/\sigma$ for 1%, 2%, and 5%. Use §5.4's prefactor for a conservative underdamped envelope bound.

2. **Fig. 3.26's shading marks the excluded region.** The text says the allowable region is to the left of the boundary. Announce it.
3. **" $\sigma \geq 4.6/3 = 1.5 \text{ sec}$ " in Example 3.27 has the wrong unit.** σ is a rate. Correct it in passing; it costs five seconds and prevents a persistent confusion.
4. **The rise-time constant is not accurate at high damping.** $\omega_n t_r$ is 2.13 at $\zeta = 0.7$ and 3.36 at $\zeta = 1$, not 1.8. Demo 2 shows a boundary case failing because of this.
5. **The RHP-zero overshoot claim needs its comparison stated** (§10.1). Against a same-distance LHP zero it depresses the overshoot; against no zero it does not.
6. **Fig. 3.34 and the written model have different final-value conventions.** Normalisation is an inference, not a stated source fact. Label $H/H(0)$ explicitly if reproducing the plot.
7. **Normalised axes everywhere in §3.5.** Figs. 3.27, 3.28, 3.36 and 3.37 all plot against $\omega_n t$. Numbers read off them must be divided by ω_n .
8. **Nonminimum phase and instability are different properties.** The 747's complex pair is stable, but its altitude integrator makes the full plant non-BIBO-stable. A clearer stable nonminimum-phase example is $(1 - s)/(s + 1)^2$; its RHP zero coexists with LHP poles.

18. Demonstration index

#	Script	Section	What it settles	Use
1	l3_demo1_step_specs.py	§5	Exact peak formulas, a rise-time fit, and an approximate settling rule compared with the full envelope bound.	slide
2	l3_demo2_spec_regions.py	§6	A pole pair on the specification boundary that fails the specification.	live
3	l3_demo3_zeros.py	§8–§10	The derivative decomposition; overshoot versus α ; Example 3.28's exact cancellations.	slide
4	l3_demo4_extra_pole_aircraft.py	§11–§12	The 747's undershoot and final value; extra-pole rise times matching Fig. 3.38.	live

19. Where this lecture sits

Lecture	Book sections	Content
L1	3.1	Convolution, transfer functions, frequency response, partial fractions, final value, poles and zeros
L2	3.2, 3.3	Block diagrams, effect of pole locations
This lecture (L3)	3.4, 3.5	Time-domain specifications; effects of zeros and additional poles
L4	3.6–3.9	Stability, Routh's criterion, system identification, scaling, history